

JGR Solid Earth

RESEARCH ARTICLE

10.1029/2026JB033768

Key Points:

- Injection protocols to enhance induced seismicity are tested
- Success of fluid preconditioning is evaluated in light of injection history and environmental effects
- Mw−0.5 mainshock led to co-seismic hydromechanical phenomena

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Gischig, V. S., Meier, M.-A., Lambiasi, A., Jiang, D., Hertrich, M., Gholizadeh Doonechaly, N., et al. (2026). Hydraulic stimulation experiments attempting to enhance induced seismicity for earthquake physics research. *Journal of Geophysical Research: Solid Earth*, 131, e2026JB033768. <https://doi.org/10.1029/2026JB033768>

Received 3 FEB 2026

Accepted 23 JUN 2026

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Hydraulic Stimulation Experiments Attempting to Enhance Induced Seismicity for Earthquake Physics Research

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Abstract Improving our understanding of induced and natural earthquakes benefits from controlled experiments in insitu laboratories. To investigate the processes during an Mw~0 earthquake, we performed the “Mzero” experiments in a densely instrumented testbed of the Bedretto Underground Laboratory. These multi-day hydraulic stimulation experiments went the opposite direction to typical induced seismicity research in that they were designed to enhance seismic rupture. In the first experiment, MzeroA, the rock mass was preconditioned by injecting water for 4 days at a pressure just below fracture reactivation pressure followed by a hydraulic stimulation above reactivation pressure. This strategy aimed at increasing the fractures area near critical stress conditions for shear failure, thus facilitating larger ruptures. During MzeroA, an event with Mw−0.54 was induced, followed by a distinct aftershock sequence. This event was one magnitude unit larger than any preceding event. In the second experiment, MzeroB, stimulation was initiated directly, without preconditioning. Compared to MzeroA, MzeroB exhibited higher seismicity rates, a larger seismicity cloud, and a pronounced migrating seismicity front. Combining seismological and hydromechanical observations, we discuss mechanisms that may have influenced the contrasting seismicity responses. In addition to fluid preconditioning, stress transfer from the Mw−0.54 mainshock, and natural seasonal pressure changes, may have contributed to modulating the seismogenic response. However, the relative importance of these processes remains uncertain given experimental limitations. Our results highlight both the potential and challenge of designing hydraulic stimulations to enhance or suppress seismic rupture, with implications for earthquake physics and induced seismicity.

Plain Language Summary Understanding how earthquakes start, grow and stop is challenging because natural earthquakes cannot be controlled and observed from close-by. Underground laboratories offer a unique opportunity to study earthquake processes through carefully designed fluid injection experiments during which earthquakes are induced. In this study, we conducted two multi-day hydraulic stimulation experiments at the Bedretto Underground Laboratory in Switzerland with the goal of increasing the likelihood of inducing a small earthquake that could be observed in detail. The two experiments differed in whether the rock mass was “preconditioned” by injecting fluid at pressures just below fault reactivation before the main stimulation. One experiment included this preconditioning phase, while the other did not. The experiments produced clearly different outcomes. The preconditioned experiment generated a step-out Mw−0.54 earthquake followed by an intense aftershock sequence and measurable changes in fluid flow and rock deformation. In contrast, the experiment without preconditioning produced higher overall seismicity rates and stronger outward migration of

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earthquakes but no similarly dominant mainshock. Detailed pressure, flow, and deformation measurements reveal that both the injection protocol and the prior history of stimulation strongly influence the seismic response. In addition, stress changes caused by the mainshock and natural seasonal variations in subsurface fluid pressure likely affected the results. These findings highlight the challenges of deliberately enhancing or suppressing induced seismicity and have implications for earthquake physics experiments and for managing seismic hazard in geoenery applications.

1. Introduction

Earthquakes remain one of the most significant natural hazards for human society. Natural earthquakes are responsible for enormous losses in infrastructure and life. Additionally, induced earthquakes from anthropogenic activity can cause significant damage and threaten future development of subsurface energy resources (Ellsworth, 2013; Grigoli et al., 2017; Moein et al., 2023; Schultz et al., 2020). Efforts to improve the understanding and hazard assessment of earthquakes have been made by studying earthquake processes using seismic and geodetic signals recorded at surface networks (e.g., Lanza et al., 2026; Schliwa et al., 2024), as well as multi-disciplinary monitoring in near-fault observatories (Chiaraluce et al., 2022) and deep boreholes crossing fault zones (Bohnhoff et al., 2017; Gupta et al., 2015; Zoback et al., 2011). In recent years, in situ experiments on faults in underground laboratories or mines have gained momentum (STIMTEC; Boese et al., 2022; Aspö; Zimmermann et al., 2019; Grimsel; Amann et al., 2018; BedrettoLab, Gischig et al., 2020; JAGUARS, Plenkens et al., 2011; EGSCollab, Kneafsey et al., 2025). These underground laboratory experiments are usually designed to support the safe development of underground geoenery applications, such as geothermal energy, CO₂- and H₂-storage, oil and gas extractions, and related waste-water disposal, etc (Zhou et al., 2024). Further, these experiments provide essential steps forward into our understanding of earthquake physics as they fill a variety of research gaps between the scale of rock-deformation experiments in laboratories (e.g., Bianchi et al., 2024; Chen et al., 2025; Salazar Vasquez et al., 2024; Selvadurai, 2019; Shi et al., 2024) and natural tectonic earthquakes (Cocco et al., 2016, 2023).

The Fault Activation and Earthquake Rupture (FEAR) project (Achtziger-Zupancic et al., 2024; Meier et al., 2026; Mosconi et al., 2025) aims to observe phenomena associated with induced Mw 1.0-scale earthquakes within extensive multi-parametric and multi-scale 3D monitoring systems at the decameter-scale at the “Bedretto Underground Laboratory for Geosciences and Geoenergies” (BedrettoLab; aka BULGG; Ma et al., 2022). The experimental controllability and accessibility for instrumentation at the BedrettoLab potentially allows us to observe details of complex on-fault processes, such as precursory phenomena and rupture dynamics in connection with fault architecture and stress conditions, enabling unprecedented insights through experimental earthquake physics. Being able to induce seismicity through targeted high-pressure fluid injections plays an important role for such experiments. It allows producing earthquakes at specified times and locations, albeit in a probabilistic sense regarding occurrence time, location and magnitude. Understanding these fluid-induced earthquakes is relevant for natural earthquakes, because (a) they help understanding earthquakes and slow faulting processes at many scales (Cocco et al., 2016; Gabriel et al., 2024; Ide et al., 2007; Shi et al., 2024), and (b) subsurface fluid migration and pressure transients have been linked to various seismic sequences and therefore, fluids must play an important role in many earthquake faulting processes to some degree (e.g., Cocco & Rice, 2002; Husen et al., 2007; Miller et al., 2004; Ross et al., 2020; Simon et al., 2025).

Given that underground laboratory earthquakes typically have magnitudes Mw << 0 (e.g., Gischig et al., 2025), experiments for earthquake rupture dynamics would benefit from injection procedures (in terms of injection pressure and flow rate as well as their duration and sequencing) that increase the chance of inducing larger ruptures, for example, at the Mw ~ 1.0 scale. This objective is contrary to most induced seismicity research, in which much effort is directed toward designing hydraulic stimulation strategies that may lead to lower levels of induced seismicity (e.g., Cappa et al., 2019; Jia et al., 2022; Kwiatek et al., 2019; Ritz et al., 2022). For instance, cyclic stimulation protocols, that is, so-called fatigue hydraulic fracturing, were tested at the laboratory and underground laboratory scale (Zang et al., 2019), but also at the full field-scale of geothermal projects (e.g., Pohang and Helsinki; Hofmann et al., 2018, 2019; Kwiatek et al., 2019). Although reduced levels of seismicity have been implied by some of these studies, more applications are required to conclusively assess their effectiveness to limit seismicity.

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Numerical studies, however, also suggest that cyclic injections may in fact produce higher levels of seismicity in terms of event rates and magnitudes (Johan et al., 2018; Kroll & Cochran, 2024). Other numerical studies find that slow ramping up of the injection could produce larger ruptures depending on the degree of criticality of the fractures (Ciardo & Rinaldi, 2022), or that fluid production from fractures prior to injection may lead to suppression of the nucleation of dynamic ruptures or arrest of their propagation (Fryer et al., 2023). For a given injected volume, longer durations may be associated with larger earthquakes (Langenbruch et al., 2024; Moein et al., 2026). Further, the reactivation of critically stressed pre-existing faults tends to produce larger magnitude events than stimulation of new fractures (Schultz et al., 2025, 2026). Taken together, these studies conclude that the results strongly depend on the local geological, structural and hydromechanical conditions (e.g., rock type, fracture inventory, fault maturity, permeability, stress field, etc.) of the rock mass (Kivi et al., 2025; Tangirala et al., 2024).

This agrees with experimental findings by Villiger et al. (2020), who compared the seismogenic response of near-identical injections into neighboring fracture systems at the Grimsel Test Site in Switzerland. Other underground laboratory experiments at the decameter scale (e.g., Boese et al., 2022; Fu et al., 2021; Kwiatak et al., 2018; Schoenball et al., 2020) and at the hectometer scale (Gischig et al., 2025; Obermann et al., 2024) observed similar variability. This diversity of the seismo-hydro-mechanical responses emphasizes the difficulty in discerning the impact of alternative injection protocols, because the effect may be masked by the strong role of local geological conditions. It further demonstrates that the seismogenic response is non-stationary during subsequent stimulations in the same interval and for different fluid volumes (despite seismogenic characteristic being normalized to injection volume). The geological conditions, the injection protocol, and the stimulation history can all affect seismicity (Obermann et al., 2024). This has also been observed at the field scale (Konstantinovskaya et al., 2021; Obermann et al., 2022) as well on the basin scale (Schultz et al., 2018).

The wide range of outcomes regarding the impact of injection protocols from numerical and observational studies highlight the need for in situ experiments in underground laboratories to provide data for model validation. In this context, we designed and performed two extended hydraulic stimulation experiments (hereafter referred to as *MzeroA* and *MzeroB* experiments) at the Geothermal Testbed of the BedrettoLab, to produce earthquakes with $M_w \sim 0.0$, recorded in a densely instrumented rock volume. The experiments are preparatory for the main FEAR experiments, which aim to induce mainshocks at the $M_w \sim 1.0$ scale (Meier et al., 2026); they are designed to test if the rock mass can be prepared (i.e., preconditioned) prior to the hydraulic stimulation in a way that the chance of inducing a seismic rupture is enhanced.

Although the desired $M_w 0.0$ goal was not achieved, several $M_w -0.5$ events and their associated seismo-hydro-mechanical responses revealed observations within the rock mass that are unprecedented at this scale, and with this level of instrumentation. In this article, we describe the injection procedures and highlight the required technological developments (Section 3). We then present and analyze the induced seismicity sequence, also in comparison to previous experiments in the same fracture network (Section 4). We present pressure and deformation time series, co-seismic hydromechanical observations, as well as the evolution of hydromechanical response of the rock mass (Section 5). Finally, we discuss if and how the two injection protocols have led to the different outcomes of the two experiments and what other factors may have played a role (Section 6). A detailed analysis of seismicity with a focus on the $M_w -0.54$ main shock (i.e., pre-event seismicity, rupture dynamics and aftershock activity) will be presented in future publications.

2. Experiment Volume, Instrumentation and Interval

2.1. Bedretto Underground Laboratory

The *Mzero* experiments were performed in the BedrettoLab, which is located in a 5.2 km long tunnel in the south of Switzerland, connecting the Bedretto valley with the Furka railway tunnel. From the southeastern entrance toward the northwest, the tunnel crosses metamorphic pre-alpine units and the Rotondo granitic body (Ceccato et al., 2024; Rast et al., 2022). The latter contains the so-called Geothermal Testbed (GTB), located in a tunnel niche 2,000–2,100 m from the entrance, beneath an overburden of about 1,000 m (Ma et al., 2022; Figure 1).

The rock volume has hosted numerous hydraulic stimulation experiments since 2020. These were performed as part of pilot projects for enhanced geothermal systems that aimed at testing different borehole drilling and completion techniques (ZoDrEx, Meier & Christe, 2023) as well as hydraulic stimulation approaches in zonally

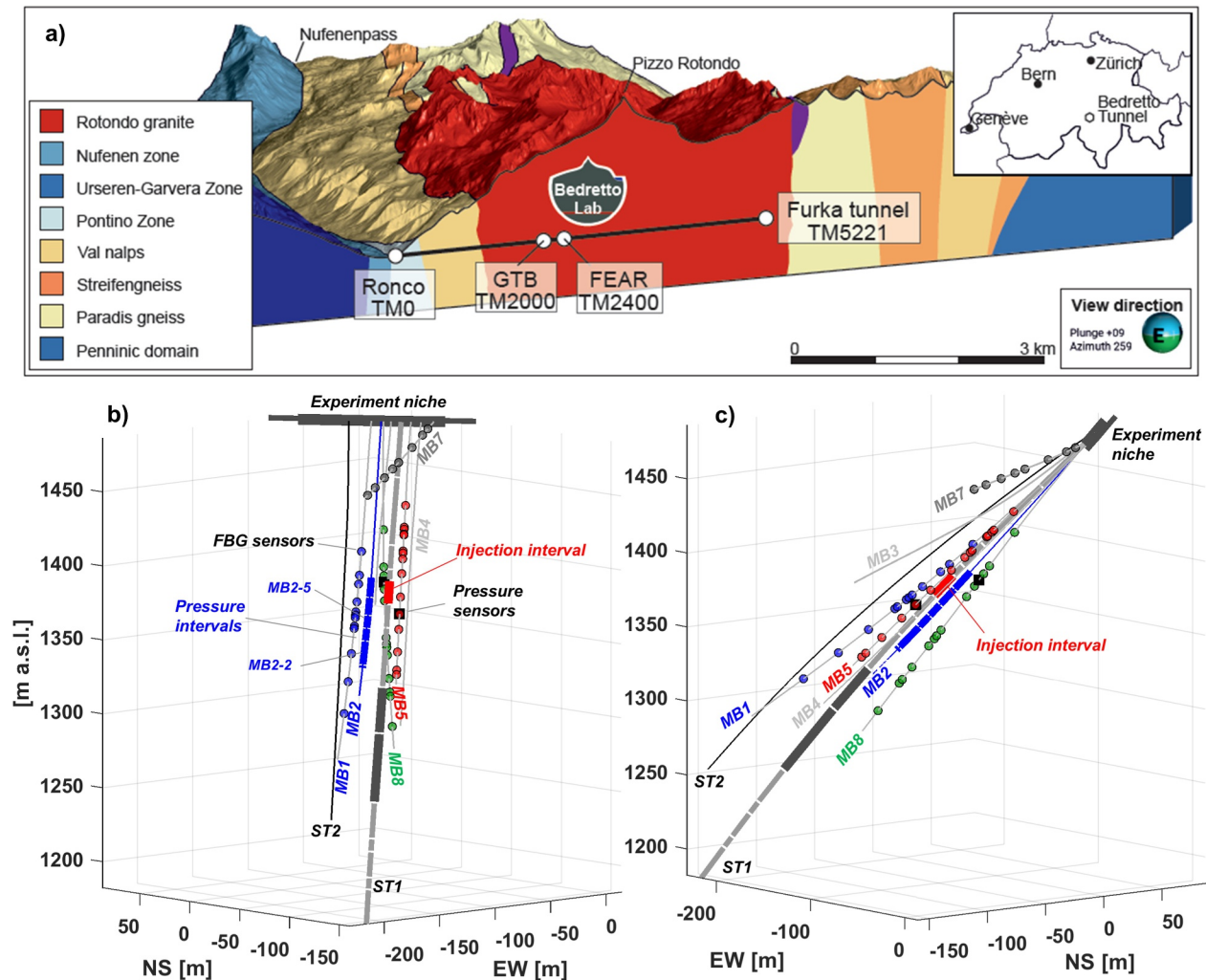


Figure 1. (a) Site overview and 3D geological model of the BedrettoLab. (Modified after Ceccato et al. (2026)). (b, c) Two view angles of the borehole setup in the Geothermal Testbed: Injection borehole ST1 contains multipacker system separating 14 injection intervals. Borehole ST2 has been completed with a steel casing in which notches were drilled with microdrilling techniques at different depths (Meier & Christe, 2023). MB2 contains a multipacker system with 7 intervals to measure pressure uphole (6 of 7 sensors functional). MB1, 4, 5, 7, 8 contain fiber-bragg grating (FBG) sensors (53 and 50 of 70 functional during MzeroA and MzeroB, respectively). MB5 and MB8 each contain two downhole pressure sensors (so-called CTPPs 1 of 2 working per borehole). During the Mzero experiments, the boreholes MB1, 4, 5, and 8 were measured with a DSS interrogator. MB3, 5, and 8 were monitored with a DAS interrogator.

isolated boreholes (DESTRESS and VALTER, Shakas et al., 2020; Giardini et al., 2022). The setup and installation of the monitoring system within this rock volume is presented by Plenkers et al. (2023). Results on induced seismicity of past hydraulic stimulations between 2021 and 2023 are presented by Obermann et al. (2024) and Roskopf et al. (2025), while hydromechanical observations are presented by Bröker, Ma, Gholizadeh Doonechaly, et al. (2024) and Golizadeh et al. (2025). Results of an extensive stress characterization effort are presented by Bröker and Ma (2022); Bröker, Ma, Zhang, et al. (2024) and Zhang et al. (2023).

2.2. Boreholes and Monitoring System

The boreholes of the GTB are shown in Figure 1. The injection borehole ST1 was completed with a multipacker system isolating 14 intervals of different lengths. The intervals are separated from each other by 1.5 m long packers that are permanently installed. The intervals between the packers can be accessed with sliding-sleeves. The injection borehole is surrounded by monitoring boreholes that host a wide range of different monitoring systems (Figures 1b and 1c for hydromechanical monitoring, Figure S1 in Supporting Information S1 for seismic monitoring).

The multiparametric monitoring system is hosted in the boreholes of MB 1–5, 7, and 8. Seismic monitoring consists of acoustic emission (AE) sensors, accelerometers, as well as geophones at the bottom of each monitoring borehole (except MB2). MB2 contains seven pressure monitoring intervals separated by a multipacker system and monitored with uphole pressure sensors. Four pressure monitoring cells (0.7 m length) equipped with downhole sensors are located in MB5 and MB8. Linear deformation (i.e., longitudinal deformation along the borehole axis) is monitored with fiber Bragg grating (FBG) sensors with a 1 m base-length in MB 1, 5, 7, and 8. All monitoring boreholes equipped with fiber-optic cables were interrogated with distributed acoustic sensing (DAS) devices and/or distributed strain sensing (DSS) devices. All monitoring boreholes and ST1 and ST2 are equipped with a distributed temperature sensing (DTS) cable. More details on the borehole-based monitoring system are given by Plenkers et al. (2023).

The experiments were additionally accompanied by frequent water sampling of microbial activity and geochemical components (Zhe et al., in prep). Samples were taken from ST2 outflow as well as from the tunnel inflow at the start of the Geothermal Testbed (TM2000). Using a portable mass spectrometer (miniRuedi; Lightfoot et al., in prep), the noble gas concentrations of the ST2 outflow water were also monitored. During MzeroA, Krypton was injected at the beginning of the main stimulation as a noble gas tracer to be monitored.

2.3. Past Hydraulic Stimulation and Target Stimulation Interval

Informed by the results of past hydraulic stimulations in the different intervals of ST1 (see timeline in Figure 2a and Figure S2 in Supporting Information S1), we chose interval 11 as target interval for MzeroA and MzeroB. The borehole depth of the interval is 132.24–150.50 m and its length is 18.26 m. It is located within the high-resolution part of the monitoring system, and it has moderate transmissivity so that the available pump circuits can achieve sufficiently high flow rates to reach required injection pressures.

The interval has already seen a number of hydraulic stimulations as can be seen in the timeline in Figure 2a. The first stimulation in March 2022 was part of the Phase 1 stimulation series of the VALTER experiment between Nov. 2021 and March 2022, during which the intervals 7–14 were tested with near-identical injection protocols, comprising two injection cycles during which pressure was ramped up in steps (step-pressure tests) and which were separated by a shut-in and bleed-off phase of a few hours. In Phase 2 of VALTER (June 2022–July 2023), certain intervals were targeted for longer stimulations (interval 8, 9, and 10 jointly), or to test alternative injection protocols (interval 11 and 12, Phase 2A and 2B each). Since these Phase 2 stimulations, interval 11 has been used as test interval for several smaller stimulations, some of which were pump system checks in preparation of the Mzero experiments.

The results of the hydraulic stimulation experiments between 2021 and 2023 (VALTER Phase 1 and 2 stimulations) are reported by Bröker, Ma, Gholizadeh Doonechaly, et al. (2024), Obermann et al. (2024) and Gholizadeh et al. (2025) and are also summarized in the Figure S2 of Supporting Information S1. During the Phase 1 stimulations with relatively small volumes (0.3–14 m³), the seismogenic response represented by the seismogenic index (a_{fb} -value, Mignan et al., 2017, Σ, Shapiro et al., 2010) and the b -value have been very variable; seismicity was most intense in intervals 7 and 13 with the largest a_{fb} -value and lowest b -value. These intervals had also the largest transmissivity prior to stimulation. In comparison, the seismogenic response in our target interval 11 was in the mid-range. Interestingly, the a_{fb} - and b -values of intervals 8, 9, 10, 11, 12 became more similar upon revisiting with larger stimulations in Phase 2, and became more comparable to those of interval 7 and 13. The largest event before the Mzero experiments had $M_w \sim -1.64$, and was recorded during the Phase 2 stimulation of interval 8, during which a volume of 273.8 m³ was injected. The evolution of further interval 11 stimulation is discussed in Section 4.

3. Experiment Design

3.1. Concept of Pressure Preconditioning

To establish an injection procedure capable of inducing a $M_w \approx 0.0$ mainshock, we estimated the volume required to encounter one such event from a statistical point of view. For this, we used the seismogenic index (a_{fb} -value) and the b -values of past stimulations of interval 11 (Obermann et al., 2024; Figure S2 in Supporting Information S1). This gave volumes of 840 m³ for the Phase 2A values ($a_{fb} = -4.18$, $b = 1.55$) and 15'000 m³ for Phase 2B values ($a_{fb} = -2.93$, $b = 1.33$). Injectivity estimates by Bröker, Ma, Gholizadeh Doonechaly,

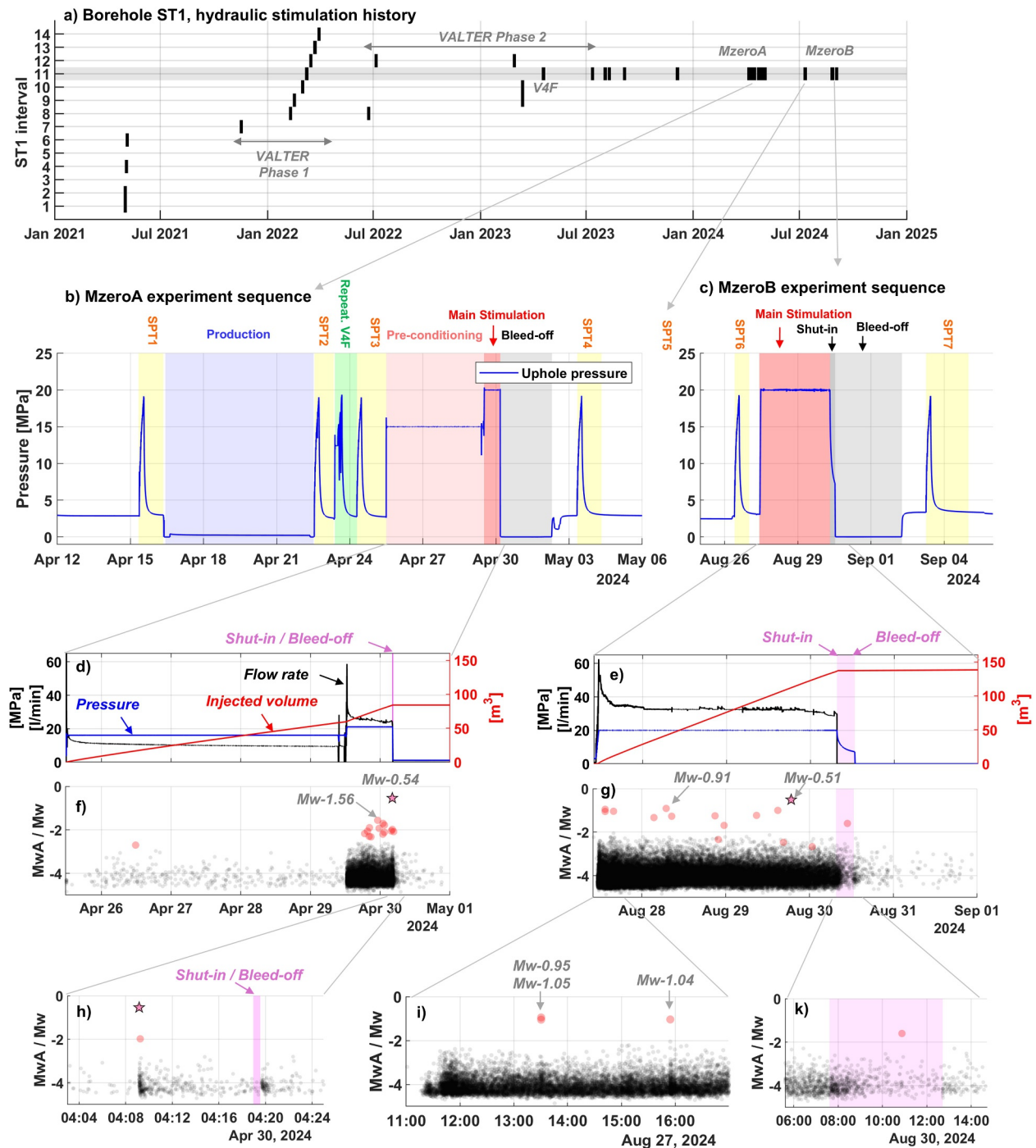


Figure 2. (a) Hydraulic stimulation history of borehole ST1. The VALTER Phase 1 and 2 stimulations are documented by Obermann et al. (2024) and Bröker, Ma, Gholizadeh Doonechaly, et al. (2024). For the stimulation in May 2021, see Gischig et al. (2025). (b) Experiment sequence of MzeroA. (c) Experiment sequence of MzeroB. (d) Flowrate, injection pressure and injected volume of MzeroA preconditioning and main stimulation. (e) Same for the MzeroB main stimulation. (f) Magnitude versus time. Note that two different magnitude scales are shown here; an apparent moment magnitude Mw (gray dots) based on AE sensors, and a real moment magnitude Mw based on geophone and seismometer data for the largest events (red filled circles). (g) Magnitude vs. time for MzeroB. (h) Close-up of mainshock and shut-in/bleed-off phase of MzeroA. (i) Close-up of first larger events during MzeroB. (k) Close-up to shut-in and bleed-off of MzeroB.

et al. (2024) of 20–30 L/min for injection pressures of ~19–21 MPa (i.e., at the pressure limits given by pump specifications) would require injection at maximum rates for a period of 20–350 days to achieve these volumes. The uncertainties are considerable owing to the variable seismogenic responses of repeat injection even in the

same interval (Figure S2 in Supporting Information S1). Nevertheless, purely statistical considerations appeared to suggest that experiments would be impractical for inducing $a \approx \text{Mw}0$ mainshock. In fact, using the above values the probability of inducing a Mw0 is estimated on the order of 3%–10%, if 100–300 m³ is injected (Gischig et al., 2025).

Beyond statistical considerations, we explored the physically motivated concept of fault/fracture hydraulic *preconditioning* to improve the likelihood of inducing a larger event. The underlying idea is to first reduce effective normal stresses across an extended fracture network via slow longer-term pressurization, in order to bring it closer to failure. Then a larger rupture extent may result, once pressure is suddenly increased to induce failure. This requires that fluid is injected at pressures close to but below reactivation pressure, also termed jacking pressure (Bröker, Ma, Gholizadeh Doonechaly, et al., 2024; Haimson & Cornet, 2003), for sufficiently long time to pressurize a significantly larger rock volume. Following this preconditioning, the pressure is ramped up quickly to maximize the chance of larger dynamic ruptures. The concept is supported by laboratory tests (Bianchi et al., 2026) and hydromechanical models (see numerical model results in, Figure S3 of Supporting Information S1). These show that a stimulation preceded by a preconditioning phase leads to a larger rupture area even though no slip accumulates during preconditioning. The concept is also supported by the findings of Ciardo and Rinaldi (2022), which point out that slow ramping up the pressure can enhance seismicity. Moein et al. (2026) also suggest that longer durations of injection may ultimately lead to larger earthquakes similarly supporting preconditioning. In principle, such preconditioning could be used to effectively control the size of the pressurized region, and hence the magnitude of the target earthquake (Mosconi et al., 2025).

We designed the Mzero experiments in two parts to test the effect of preconditioning: MzeroA, with an extensive (4.5 days) preconditioning-phase before the high pressure stimulation, and MzeroB, without preconditioning before stimulation. Additional shorter injections and production tests were part of the overall experiment sequences. The experiment sequences are presented in Figure 2 and described in detail in the following.

3.2. Injection Procedure

Hereafter reported pressure values are measured uphole (i.e., in the tunnel) unless stated otherwise. *Preconditioning* implemented during *MzeroA* included continuous injection with a constant pressure of 15 MPa for 96 hr. During this phase, 59.5 m³ of water was injected. Preconditioning was followed immediately by the main stimulation involving injection at 20 MPa that was planned to last up to 4 days. The main stimulation phase was terminated after 16 hr, only a few minutes after the occurrence of the Mw=0.54 main shock. This followed the predefined halt criteria (see Section 3.3) stating that the stimulation is terminated when experiment goals are reached. Note that the initial manual estimate of the mainshock magnitude was MLC=0.2, considered within the error margin of an Mw0, thus leading to injection halt. A total of 24.6 m³ was injected during the 20 MPa main injection, summing up to a total of 84.1 m³ for the entire *MzeroA* experiment.

The *MzeroB* experiment was performed 4 months after *MzeroA* (Figure 2c); injection included only a main stimulation at a constant 20 MPa injection pressure that lasted 70 hr. Specifically, no preconditioning was performed prior to the main stimulation. During the *MzeroB* experiment, a total volume of 146.5 m³ was injected.

Prior to *MzeroA*, a series of additional experiments were performed. A *production test* was performed to investigate production-induced poroelastic effects in the surrounding fracture network. The scope was to investigate whether undrained poroelastic unclamping of the fracture network around Interval 11 enhances its seismogenic response, or if a drained response leads to a lower seismogenic response. For this, 543 m³ of water was produced during ~6 days from the very transmissive Intervals 6 and 7. Further, a shortened version of the preconditioning concept was performed in interval 11 to repeat a nearly identical test in April 2023 (i.e., Phase 2A test, Figure 2a, also termed VALTER for FEAR (V4F) test). Preconditioning at 13.5 MPa was done for 2 hr, followed by three stimulations at 15, 20, and 30 L/min each lasting 1 hr and separated by a 0.5-hr shut-in phase.

Precise knowledge of the reactivation pressure was essential to maximize the pressure during the preconditioning phase. For this reason, the so-called step pressure tests (SPT) were performed. They were not only performed before preconditioning, but also after and between different components of the entire experimental sequence to keep track of the evolution of injectivity below and above reactivation pressure (SPT1–SPT7, Figure 2b). Thus, SPT1 serves as the baseline observation; SPT2 was performed after production and SPT3 after the repeat V4F test and before the actual *MzeroA* experiment. These three SPTs gave a robust estimate of the jacking pressure and

thus the maximum preconditioning pressure. SPT4 was done a few days after MzeroA and SPT5 2 months after MzeroA. A day before and some days after MzeroB, SPT6 and SPT7 were performed.

It is important to note that the SPTs were all performed in an identical manner. This was possible because of the novel remote-controlled pump circuit system that was developed for these and future experiments at the BedrettoLab. The remote-controlled system allowed for running predefined injection protocols in a highly reproducible form, both in pressure- or flow-controlled mode. This was exploited for the SPT system to be able to detect subtle changes in the hydromechanical response (jacking pressure, injectivity, seismogenic response, etc.) of the fracture network.

3.3. Safety Aspects

The remote-controlled pump system is also a key element in the safety concept for our experiment (and future experiment of the FEAR project). It facilitates the remote-controlled steering of all injection pumps, as well as shut-in or bleed-off procedures, so that no personnel need to be in the tunnel during experiments. This safety concept is connected to a traffic light system based on ground motion and magnitude thresholds, to anticipate and respond to unacceptable levels of seismicity (e.g., Gischig et al., 2025): a magnitude $ML_c \geq 0.0$ or a ground motion $PGV > 2.5$ mm/s implies a yellow alert triggering the shut-in of the borehole and reconsideration of the experiment; a magnitude $ML_c \geq 0.5$ or a ground motion $PGV > 15$ mm/s implies an orange alert triggering the shut-in and bleed-off of the borehole and halt of the experiment. Note that the magnitude thresholds rely on ML_c estimates, a calibrated local magnitude estimate, which is what the BedrettoLab seismic background network provides automatically (Mesimeri et al., 2025).

4. Seismological Observations

4.1. Earthquake Production Rates and Magnitudes

Time series of injection rates and pressure, along with the temporal seismicity evolution are shown in Figures 2d–2g for both MzeroA and MzeroB. The reported (apparent) moment magnitudes M_wA rely on the magnitude scale developed for the AE-sensor network (Obermann et al., 2024). These are amplitude-based magnitudes derived in the time-domain and adjusted to realistic moment magnitude levels using records of calibrated borehole accelerometers. For the largest events that were detected on calibrated geophones at the bottom of the boreholes and at tunnel stations, a moment magnitude M_w was calculated with the SPAR algorithm by Supino et al. (2019). Note that for events recorded at the geophones and tunnel sensors, a local magnitude ML_c is computed in near-real time (Mesimeri et al., 2025). Contrary to the previous seismic data acquisition with software Dugseis (Roskopf et al., 2024), SeisComp was used (Mesimeri et al., 2025). The seismic monitoring procedure is described in detail with the seismicity catalog release (Scarabello et al., 2025).

During preconditioning of experiment MzeroA, injection pressure was held at 15 MPa pressure. The flowrate was initially at 17.5 l/min, but then decayed to a steady flow rate of 9.5 l/min. A total number of 328 seismic events were detected corresponding to an average event rate of 3–4/hr. During the main stimulation at 20 MPa, the earthquake production rate initially increased to >20 events/min and then quickly reduced to an average of about 7 events/min. Flowrate during this phase started at 35 l/min and decayed to 24 l/min (Figure 2d). A tendency toward larger magnitudes was observed during the main stimulation, but magnitudes stayed below $M_w-1.56$. An exceptional event with $M_w-0.54$ ($ML_c-0.2$) occurred on 30 April 2024 04:09:09 UTC after almost 16 hr of stimulation. The event—hereafter referred to as the MzeroA mainshock—was followed by intense aftershock activity with up to 15 detected events/second (Figure 2h). The continuous waveforms revealed that the automatic seismicity analysis algorithms could not properly process all events during this sequence, because the rate was so high, which implies that the aftershock activity was even more intense than reported (Meier et al. in prep). The MzeroA mainshock stands out in that it is at least one magnitude larger than the second largest event during MzeroA.

The fast manual re-estimate of $ML_c-0.2$ was close (i.e., within the magnitude uncertainty) to the experiment goal of reaching a $ML_c0.0$. Therefore, the experiment was halted by remotely triggered shut-in of the well (i.e., stop the injection) at 9 min 44s after the mainshock occurred. Bleed-off was initiated only 42 s later. Bleed-off was again associated with a pronounced increase of seismicity rates (max. 3 events/second, Figure 2h).

During MzeroB, injection pressure was ramped up to 20 MPa with short steps within 50 min, without any hydraulic pre-conditioning. In the ramp-up process, flow rates reached a peak of 62 l/min. Then the flow rate rapidly decayed to 50 l/min and was followed by another increase, reaching its peak of 53 l/min about 20 min after the first peak (Figure 2e). During this phase, the seismicity rate increased from about 80 to 100 events/min, that is, 4–5 times the rates observed early in the MzeroA main stimulations. During the following 4 hr, seismicity rates remained high at 50 events/min. A doublet of larger, nearly collocated events (M_w –1.05 and M_w –0.95, with <1 s in between) as well as a separate larger event with M_w –1.04 occurred during this time. These three events were accompanied by distinct aftershock sequences (Figure 2i), albeit not as intense as the sequence after the MzeroA mainshock. After this phase, the seismicity rate gradually decreased to about 10–12 events/min at the end of the 70 hr stimulation. Up to that point, the flow rate reached a steady state rate of about 33 l/min. The two largest events of MzeroB with magnitude M_w –0.91 and M_w –0.51 occurred 21 and 56.75 hr after the start of injection, respectively. Only during the last 4 hr of stimulation, a further drop of flow rate occurred to about 29–30 l/min. This drop was associated with a slight decrease in seismicity rate to about 8 events/min. Shut-in of the borehole was associated with a short-lived increase of seismicity to >10 events/min, which quickly decayed to 1–2 events/min. Unlike MzeroA, the borehole was bleed-off only after about 5 hr after shut-in to observe pressure and seismicity decay after shut-in. Bleed-off was followed by another short increase of seismicity with rates of about 10 events/min.

The average event rate of the two main stimulations (MzeroA: 440/hr, MzeroB: 640/hr, including all located events) is also compared by considering that the injection rates during both experiments were different. For this, we analyze the number of events per m^3 injected water (Figure 3), using only events with magnitudes larger than an overall magnitude of completeness $M_c = -4.25$. The difference is less pronounced than event rates per time with 160/ m^3 during MzeroA and 210/ m^3 during MzeroB. However, the event rate during MzeroB reaches >900/ m^3 and then decays to rates comparable to MzeroA after more than a day. Short-lived increases in seismicity rate follows the M_w –0.95 and –0.91 events. During MzeroA, the rates are more constant with a maximum of 250/ m^3 . The Gutenberg-Richter b -value is higher during MzeroA ($b = 1.30$) than during MzeroB ($b = 1.16$). Both b -value and event rate per m^3 are also resolved in more detail for the main stimulations in Figures 3a and 3b. During both injections, the b -value is higher during early times and then decreases with time.

For reference we also show the frequency magnitude distributions of all interval 11 stimulations, including past experiments from before Mzero (Figures 3d–3f). The b -values seem to vary non-systematically between individual experiments, in a range of approximately 1.2–2.0. We note a comparatively minor change in b -values between MzeroA (1.31) and MzeroB (1.17), that is, with and without preconditioning, and with strongly differing injection durations and volumes. The distributions also show that the M_w A magnitude scale is truncated at magnitudes above $M_w \approx -2.0$ because the AE sensors clipped for these events. More refined magnitude analysis will be part of future work.

4.2. Spatial Evolution of Seismicity

The seismicity cloud is shown in Figure 4. The hydraulic stimulations in interval 11 define several NE–SW (30–40°) striking sub-parallel structures (see animated GIFs in Supporting Information S1). The view in Figure 4 is perpendicular to these structures to highlight the migration of seismicity with time.

The first panel (Figure 4a) summarizes all hydraulic stimulation before the Mzero experiments. The progression of these stimulations produced seismicity that migrated outwards, reaching larger distances with each repeated experiment, but fewer events in the region that were active during the preceding stimulation (Kaiser effect). The tests SPT1, SPT2, V4F, SPT3 before the preconditioning of MzeroA (blue dots, Figure 4b) reactivated small structures close to the borehole, but also the upper region of the main structure (corresponding to the southwest part of the seismicity cloud). Interestingly, little outward migration of seismicity is observed during the succession of these stimulations; seismicity immediately reaches the mentioned upper part of the previously active region. Also, during preconditioning of MzeroA (orange dots, Figure 4b), only this upper region is active, while around and below the injection interval almost no seismicity is induced. The main MzeroA stimulation (Figure 4c) induces seismicity in the entire volume, that was already active in preceding stimulations. Little outward migration from the injection point with time (or injected volume as color-coded) is

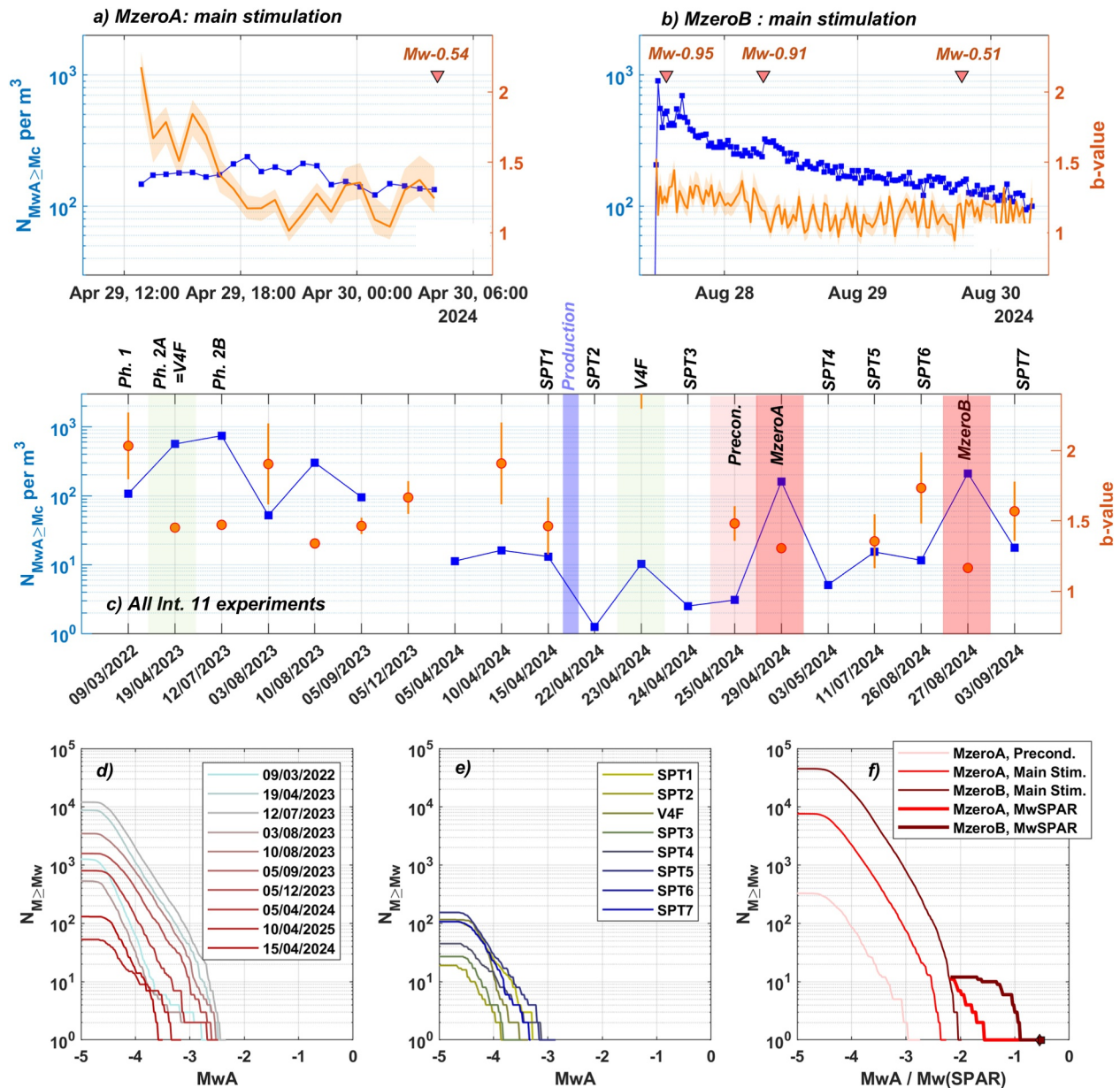


Figure 3. (a) Gutenberg-Richter b -value of the MzeroA main stimulation as well as the number of events per m^3 water injected. Note that both rate and b -value are computed for each $1 m^3$ injected. The b -value was only computed if 100 events were available. (b) The same as (a) for MzeroB. (c) Gutenberg-Richter b -value and event rate per m^3 injected for all interval 11 stimulations. No values could be computed if injected volume in not known (05/12/2023) or if the number of events was <100 , which is too low to compute a reliable b -value. (d) Frequency magnitude distributions (FMD) of seismicity catalogs from injection prior to the Mzero experiment sequences. (e) FMD of small volume injection of the Mzero experiment sequence. (f) FMD of the preconditioning phase and main stimulations of Mzero and MzeroB. FMDs of both the Mw -based catalogs as well as of the subset of events, for which an Mw was computed based on the SPAR algorithm (Supino et al., 2019).

observed. Clearly, more seismicity is induced in the upper region that was activated during preconditioning. Seismicity in the lower parts was less dense and tends to occur earlier. Toward the NE, seismicity is sparser and more diffusive; little seismicity was induced here in earlier experiments. It is noteworthy that the most active upper part of the seismicity cloud is separated by a clear boundary from the inactive region above. It is at this boundary that the $Mw-0.54$ mainshock occurred. The aftershock seismicity produced by the main shock clusters around the margins of the event (Figure 4d). Seismicity associated with bleed-off is more diffusively distributed within the volume active so far, with much occurring closer to the injection interval (black dots, Figure 4d).

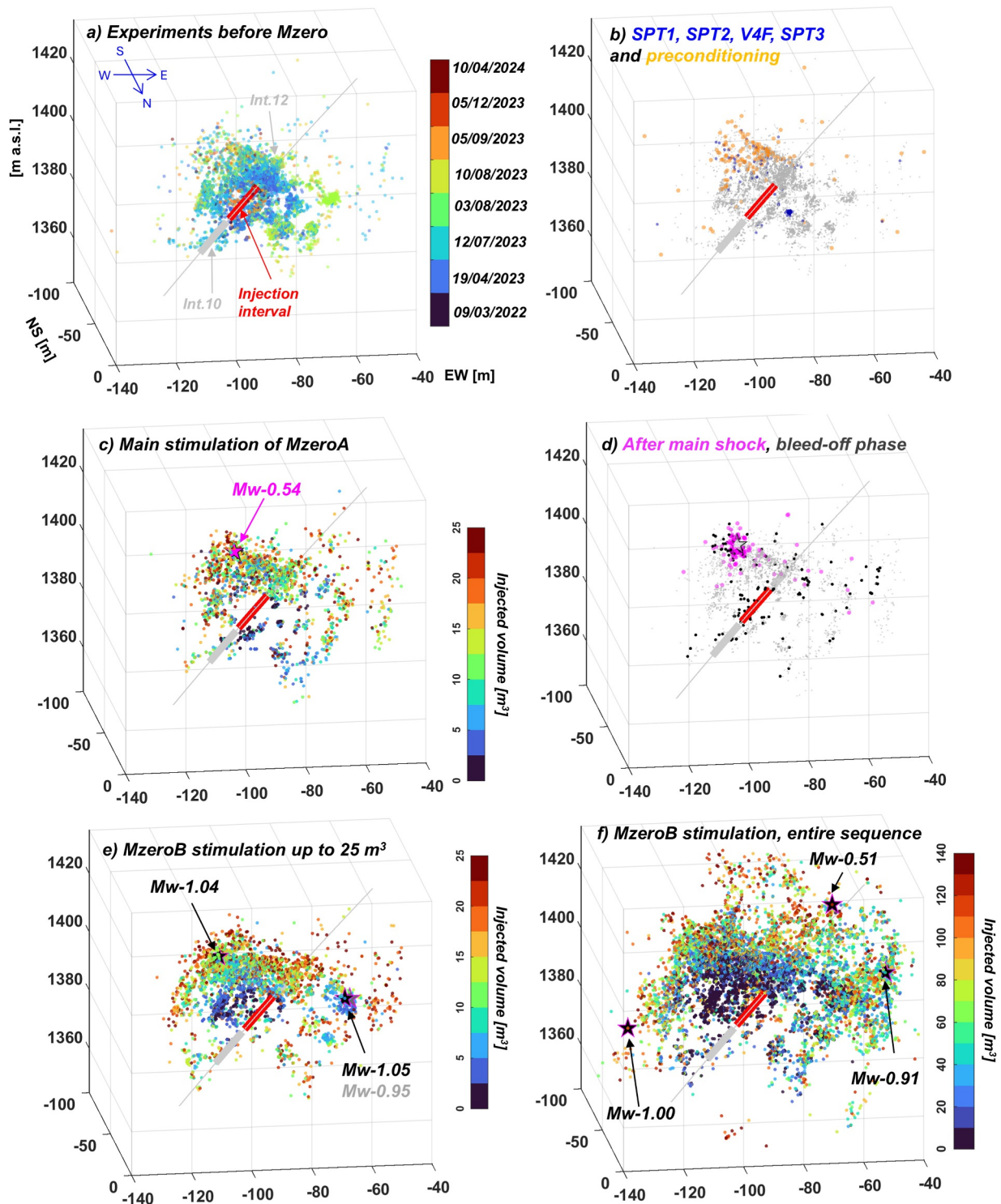


Figure 4. (a) Seismicity migration pattern of experiments before the Mzero experiments. Each color represents one of the experiments. The view angle is toward NW normal to the largest extent of the seismicity cloud. (b) Seismicity patterns of the SPTs and V4F tests as well as of the preconditioning prior to the MzeroA main stimulation. The gray dots are the seismicity recorded prior to these tests (e.g. the seismicity in (a)). (c) Seismicity pattern of the MzeroA main stimulation. The color represents the cumulative volume injected up to a certain time. (d) Seismicity recorded after Mw-0.54 main shock up to bleed-off of the well (pink). Seismicity recorded after bleed-off until the well was shut-in again. (e) Seismicity of the MzeroB main stimulation up to 25 m³ injected volume for comparability with MzeroA in (c). (f) Seismicity during the entire MzeroB injection.

The initial seismically very active phase of MzeroB is shown in Figure 4e, color-coded with injected volume up to 25 m^3 , that is, the same as the total injected volume during the MzeroA main stimulation for better comparison (Figure 4c). Up to this time (or volume), the same region as in the MzeroA main stimulation is active, however, with more seismic events and a clearer propagation direction away and upwards from the injection interval. Further, a patch toward the NE (i.e., right in the figure) was more active during MzeroB but not in MzeroA. Here, the double event mentioned in Figure 2i ($M_w - 0.95/-1.05$) was induced. The other large event during this phase ($M_w - 1.04$) was induced close (3.2 m distant) to the main shock of MzeroA. After more than 25 m^3 (i.e., more than during the MzeroA main stimulation) was injected, the seismicity cloud propagated beyond the prior extent of the seismicity cloud, upwards and toward NE. The three largest events of MzeroB (with magnitudes $M_w - 0.91, -1.00$ and -0.51) occurred close to the periphery of this extended seismicity cloud.

Close-up figures in the Figure S4 of Supporting Information S1 highlight that the structures illuminated by the seismicity align with NE-SW trending structures mapped at the boreholes wall. Although most seismicity is concentrated along one main structure, there are many subparallel structures belonging to a complex interacting fracture network. The main structure intersects the borehole above interval 11 at the packer separating interval 11 and 12 and can be correlated with a mapped structure oriented $340^\circ/50^\circ$ (dip direction/dip). Other minor seismicity structures seem to align with structures in intervals 10 and 12, that is, seismicity has been induced also close to the intervals above and below the target interval 11.

To understand the MzeroA and MzeroB seismicity evolution in the context of the entire interval 11 stimulation history, the vertical coordinate of seismicity versus time is shown in Figure 5, which also shows the injection pressure time series and the injected volume per stimulation. A vertical upward propagation pattern is observed for the VALTER Phase 1 and 2 stimulations (Figure 5a). The very first stimulation produces seismicity close to the packer, whereas the subsequent stimulations produce seismicity further away, both upward and downward from the injection interval. After the first three stimulations, seismicity does not expand much further. From SPT1 to preconditioning (Figure 5b), seismicity occurs in patches at the altitude of the lower packer, and scattered around $1,400 \text{ m}$, which corresponds to the active upper region in Figure 4b. Little vertical propagation occurs during MzeroA main stimulation as shown in Figure 4b. After MzeroA, the SPT4, 5 and 6 produce seismicity in the region around $1,400 \text{ m}$, where the $M_w - 0.54$ mainshock and its aftershock sequence occurred (Figure 4c). Unlike MzeroA, during MzeroB seismicity initiates close to the packers and propagates outwards in a similar manner as during the very first three stimulations in the interval. This time, seismicity propagates upwards beyond the previously active region (up to $1,460 \text{ m}$). Note that seismicity after shut-in as well as seismicity of SPT 7 occurs predominantly close to the interval.

Figure 5 also helps to visualize productivity in terms of seismicity during the progression of stimulations. During the stimulations prior to the Mzero sequences, the seismicity productivity “depletes” with the repeated small-volume ($1.8\text{--}8.7 \text{ m}^3$) injections. In other words, after the intense seismic activity during VALTER Phase 2 stimulations, the seismic activity decreased despite comparable injected volume. Certainly, the event rate per m^3 also depends on the injection schedule, for example, on how much volume is injected below reactivation pressure, but the tendency seems much stronger and systematic as can be explained by injection schedule only. In Figure 3c, productivity is quantified by the number of events ($M_w > M_c = -4.25$) per m^3 injected water. We observe a decrease in event rates prior to the MzeroA main stimulation followed by an increase in event rates after MzeroA.

During the nine stimulations prior to the Mzero experiments, the event rates drop from >700 per m^3 to ~ 10 per m^3 . Note that for these stimulations, the injection protocol varies. On the other hand, the SPT tests and part of the Mzero experiment sequence can be compared, because the exact same injection schedule was used. Compared to SPT1, the event rate further decreases after the production test (SPT2). Before the preconditioning, it remains on a low level (SPT3) but then increases between MzeroA and MzeroB (SPT4–6) and is largest after MzeroB (SPT7). A direct comparison of the event rate is also possible for the V4F test and its repetition during the MzeroA sequence, because they follow the same injection protocol: the event rate is an order-of-magnitude lower, with 566 events per m^3 for V4F in 2023 and about 10 per m^3 for the repeated V4F test in 2024. Note that the b-value does not show a simple systematic trend, and ranges from 2.35 to 1.05 (Figure 3).

Summarizing the seismological observations, we state that the MzeroA and MzeroB main stimulations were different in many aspects, even though the injection pressure was the same.

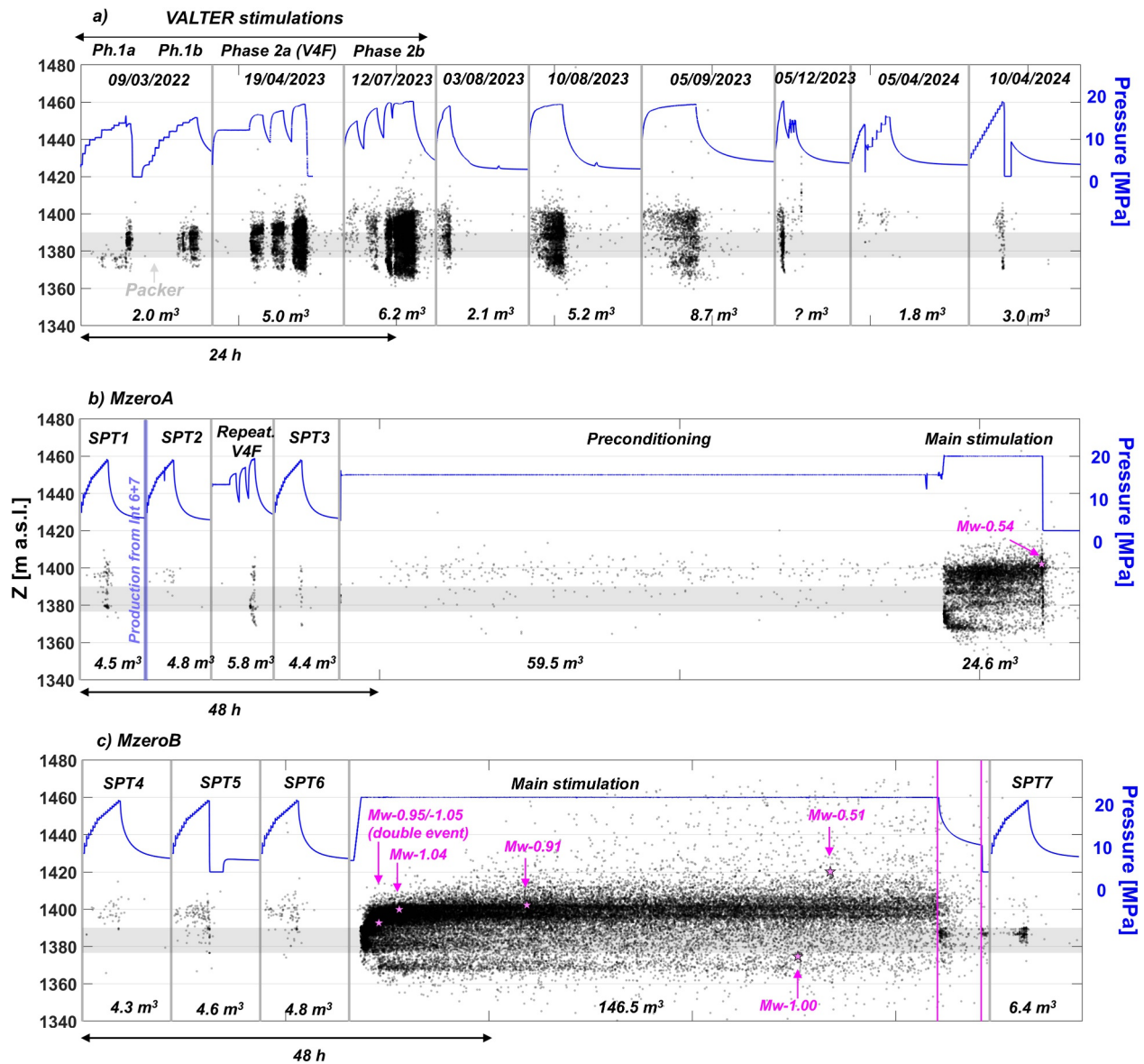


Figure 5. Evolution of seismicity over the course of all experiments performed in interval 11. The y-axis is the altitude of the seismic event locations (left axis), as well as the pressure (right axis). The x-axis corresponds is time in hours. (a) Stimulation experiment before the Mzero experiments. (b) The MzeroA experiment sequence. (d) MzeroB Experiment sequence including the SPT4-5 between MzeroA and MzeroB.

- The mainshock of MzeroA is more than a magnitude larger than the second largest MzeroA event and was associated with intense aftershocks (Figure 2). In contrast, during MzeroB, several larger events were induced but only a few of them had aftershock sequences.
- MzeroB was more seismogenic in terms of event rate per time or per injected volume compared to MzeroA, in particular in the first 1–2 days of the injection. MzeroB's event rate per time (or per injected volume) was significantly more non-stationary as compared to MzeroA (Figure 3).
- MzeroB exhibits a clear migration pattern away from the injection interval, which was not the case for MzeroA (Figure 5).

In the following sections, we explore hydromechanical observations to support the interpretation of these differences.

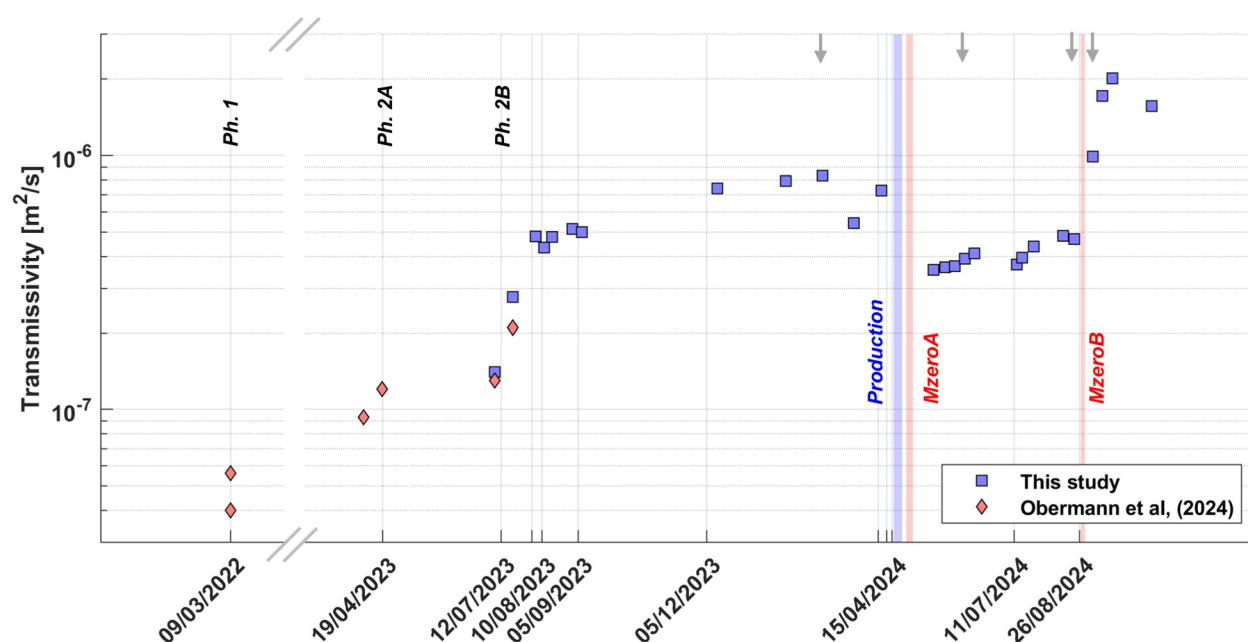


Figure 6. Evolution of hydraulic transmissivity from drawdown tests with a constant production rate of 0.75 l/min. Gray arrows indicate tests for which diagnostic plots are shown in Figure S5 of Supporting Information S1.

5. Hydromechanical Observations

5.1. Transmissivity and Near-Wellbore Flow

Hydraulic transmissivity of the injection interval has been inferred from a series of hydraulic tests between 2022 and 2024, typically before and after most stimulations. During these drawdown tests, a constant 0.75 l/min was produced for at least a day. The transmissivity evolution is shown in Figure 6, where also transmissivity values from Obermann et al. (2024) are included. Initial transmissivity was $5.6 \times 10^{-8} \text{ m}^2/\text{s}$ and reached a maximum of $1.6 \times 10^{-6} \text{ m}^2/\text{s}$ after the MzeroB sequence. The stimulations between March 2022 and August 2023 led to a gradual increase of transmissivity, which plateaued at 5 to $9 \times 10^{-7} \text{ m}^2/\text{s}$ before the Mzero experiments. After the MzeroA experiments, transmissivity dropped by a factor of two, which is unexpected given that stimulation usually leads to a transmissivity enhancement. After MzeroA, transmissivity stayed relatively constant until MzeroB, after which transmissivity increased again, by at least a factor of 5 ($2 \times 10^{-6} \text{ m}^2/\text{s}$).

The diagnostic drawdown versus time plots of four representative hydraulic tests (Figure S5 in Supporting Information S1) show that the early time of pressure depletion representative of the near-wellbore conditions follow a slope close to 0.25 for tests before MzeroA and between 0.25 and 0.5 for tests between MzeroA-MzeroB and after MzeroB. A slope of 0.25 indicates bilinear flow, characterized as fracture flow and the surrounding rock mass contributing flow to the fracture flow (Bourdet, 2010; Renard et al., 2009), whereas a slope of 0.5 indicates linear flow, that is, fracture-dominated flow with little flow from the surrounding rock mass. The change from a bilinear flow regime (before MzeroA) to a linear flow regime (after MzeroA) corroborates with the decrease in transmissivity. At the later time of the hydraulic tests, the diagnostic plots show a constant pressure derivative, which is interpreted as an infinitely acting radial flow reservoir, for all tests, except the last test (i.e., after the MzeroB experiment). The test after MzeroB shows an increase in the pressure derivative (i.e., after 4,000 s) with a slope of 0.5, which indicates a flow boundary. The reason why this was not visible in the previous tests may be the increased transmissivity was needed to make this boundary visible earlier during the test; the boundary may also have existed in the earlier tests but may not have been reached during the time of tests. A possible explanation for this apparent flow boundary may be the open borehole ST2 (about 35 m distance from ST1), that produces a trough of pressure depletion around it, effectively acting as flow boundary.

Changes in the near-wellbore hydraulic properties are also reflected in the flow field inferred from distributed temperature sensing (DTS) throughout the experiment sequences. Figures 7a and 7b show temperature profiles

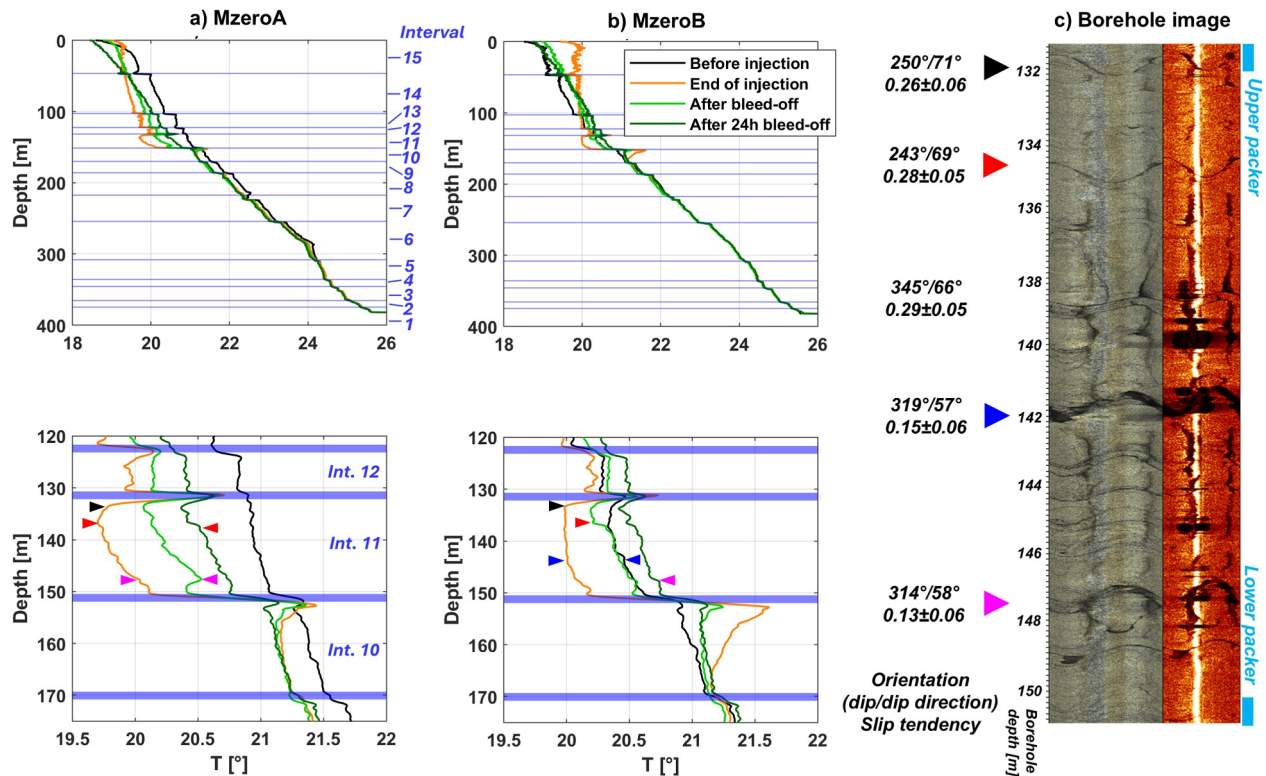


Figure 7. Temperature profiles derived from DTS along the injection borehole ST1 at different times before, during and after stimulations. Blue lines indicate packer depths. (a) MzeroA. (b) Mzero (c) ATV and OTV image of interval 11 in ST1. Indicated are open fractures for which slip tendency was computed based on orientations and the stress estimates by Bröker, Ma, Zhang, et al. (2024).

along the entire ST1 boreholes and include a zoom on interval 11 and its neighboring intervals. Note that the water injected into interval 11 is colder than the ambient rock temperature. Changes in the temperature gradient signify outflow points, that is, locations where flow enters the fracture network. The temperature at the end of the main stimulation of MzeroA (i.e., the time when temperature changes should be maximum) shows two clear outflow points: one at about 132 m, that is, very close to the upper packer, and one at 134 m depth. The outflow locations correlate with fractures observed in acoustic (ATV) and optical (OTV) televiwer logs. After opening the well, water that was pushed into the fractures flows back into the well, producing temperature anomalies. Such a distinct inflow location is seen at 148 m, where warmer water flows into the cooled interval (i.e., water from greater depth). Another one becoming visible 24 hr after bleed-off corresponds to the outflow at 134 m. These fractures can also be associated with temperature signatures during MzeroB. However, the temperature profile at the end of the injection is different, in that it exhibits nearly constant temperature between 132 and 142 m, implying that there must be another outflow location at 142 m. This outflow location also produces a step in the temperature profile before the MzeroB experiment. The actively flowing fracture at 132 m is close to the main structure of the seismicity cloud, which intersects the borehole above the upper end of the interval (Figure 4c, Figure S4 in Supporting Information S1). During MzeroA, there is also a cluster of seismicity occurring close to the fractures at 148 m. However, this cluster is no longer seismically active during MzeroB (Figure 4e, Figure S4 in Supporting Information S1). This is potentially related to the changes in near-wellbore flow regime observed in the hydraulic tests and in the temperature profiles. Changes in flow paths between repeated hydraulic stimulations are not uncommon and have been observed in other stimulation experiments (e.g., Krietsch et al., 2020).

The temperature profiles are not only perturbed in interval 11, but also above interval 12 and below interval 10. For interval 12, this is expected, as the colder injection water cools the casing above the injection interval. However, compared to the degree of cooling in the adjacent intervals 13 and 11, the temperature seems elevated, suggesting that warmer water than the injection water enters the interval through the fracture network. In interval 10, the temperature profile is also perturbed, even though the injection water leaves the borehole above it. During MzeroA, there are higher temperature anomalies at 152 m just below the upper packer and at 155 m. During

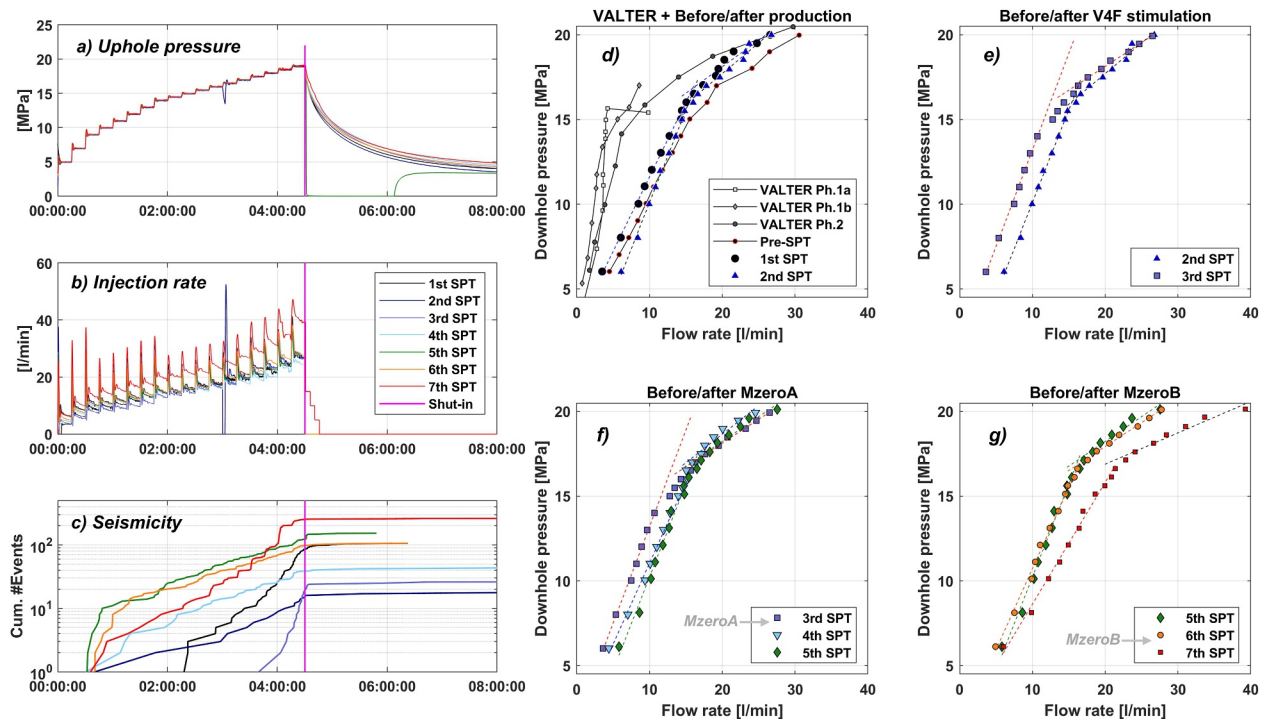


Figure 8. Summary of step-pressure tests (SPT). (a) Injection pressure of all seven SPTs. The pink line indicates shut-in of the well. (b) Injection rate during the seven SPTs. (c) Cumulative number of seismic events. (d–g) PQ-curves: flow rate versus downhole pressure at the end of each pressure step (i.e., average values of the last 1 min of each 15 min step). In (d) also the VALTER Phase 1 and 2 stimulations are included (Bröker, Ma, Gholizadeh Doonechaly, et al., 2024).

MzeroB, the temperature anomaly reaches deeper to about 165 m. The maximum of the temperature anomalies correspond to the temperatures found in interval 9 and 8. It is known from earlier stimulation that intervals 8, 9, and 10 are connected (Gholizadeh et al., 2025). Apparently, the stimulation of interval 11 pushes water from greater depths through the fracture network into intervals 10 and 12. The observations also corroborate with the details of the seismicity cloud, which shows that smaller seismically active structures correlate with fractures mapped in intervals 10 and 12 (Figure S4 in Supporting Information S1).

5.2. Hydromechanical Changes From Repeated SPT Tests

The seven identical SPT tests allow us to track changes in the hydraulic and stress conditions. Owing to the remote-controlled pump circuits, the pressure-controlled SPT tests could be conducted in a highly repeatable manner (Figure 8a). Based on pressure steps and the near-steady-state flowrate at the end of each 20 min step, we derive pressure-flowrate curves (“PQ-curves” in Figures 8d–8g), that show changes in injectivity (i.e., slope of the PQ-curve) and in the jacking pressure (kink in the PQ-curve) between the different tests. For each SPT test, injectivity was derived as the best fit line of the first seven pressure steps having injectivity below jacking pressure and of the last seven pressure steps having injectivity above jacking pressure, whereas jacking pressure was defined at the intersection of the two best fit lines.

We observe in Figure 8d and in the summary Figure 9b, that injectivity of the SPTs before MzeroA has doubled compared to 2022/2023 injections (Ph.1a, Ph.1b, Ph.2, Bröker, Ma, Gholizadeh Doonechaly, et al., 2024), which occurred in the same interval (Figure 8d), which corroborates with the increasing transmissivity between the 2022/2023 and the 2024. Before and after MzeroA, the injectivity did not change substantially, but increased clearly after MzeroB (Figure 9b). This is reflected in the volume injected up to the 12 and 20 MPa steps; these values serve as proxies for how easily water can be pushed into the formation below and above the jacking pressure (Figure 9a). Only after the MzeroB experiment did the injected volumes significantly increase by about 30%–50%, implying that the fracture network has become more permeable. Again, this qualitatively agrees with the increased transmissivity after MzeroB (Figure 6). However, the drop in transmissivity after MzeroA is not reflected in these observations on injectivity.

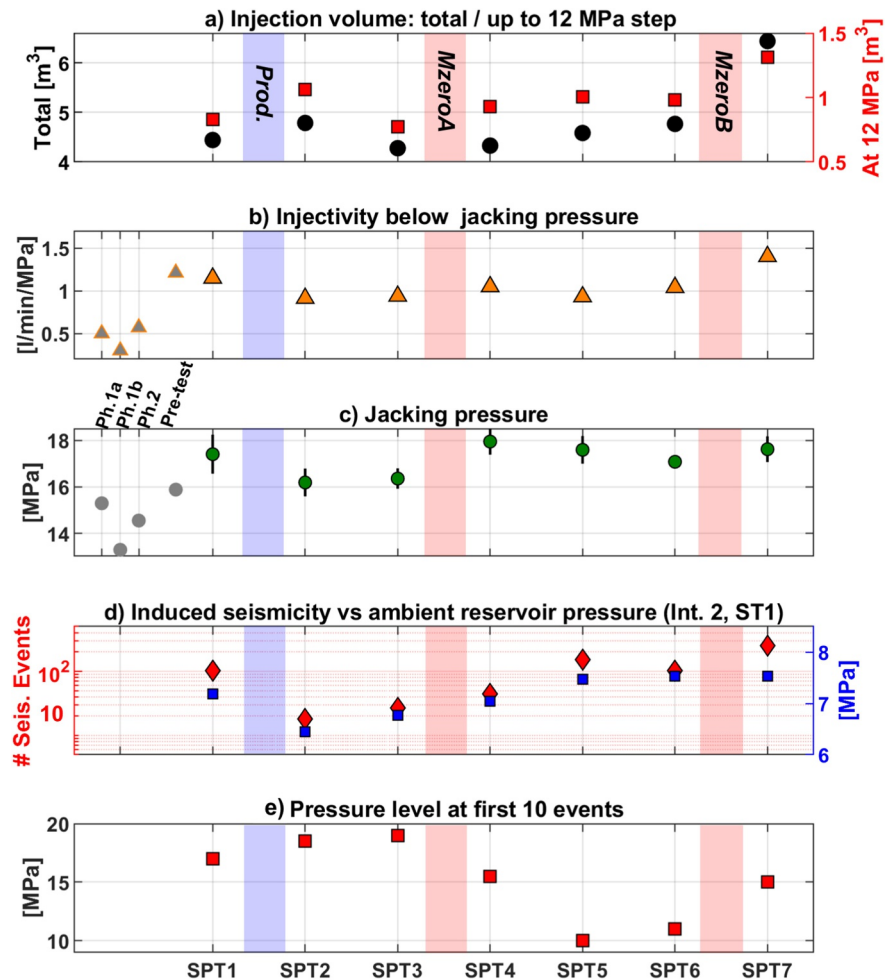


Figure 9. Summary of observables from repeated step pressure tests (SPT). (a) Total volume injected during the SPTs up to the end of the 20 MPa step (left axis), and volume injected up to the end of the 12 MPa step, that is, below the jacking pressure (right axis). (b) Injectivity below jacking pressure derived from the PQ diagrams (see Figure 8). Also included are estimates by Bröker, Ma, Zhang, et al. (2024) for the test during the VALTER stimulations (Ph.1a, Ph.1b, Ph.2; gray symbols). (c) Jacking pressure derived from the PQ diagrams. (d) Number of seismic events per SPT test and ambient pressure level from Interval ST1-2 (see Figure S6 in Supporting Information S1) at the beginning of each SPT. The correlation coefficient between seismic events per SPT and ambient pressure level is 0.93. (e) Pressure level at occurrence of the first 10 events of the seismicity sequence as proxy for the seismic reactivation pressure.

Jacking pressure shows a clear jump from 16 MPa before MzeroA to 17.6 MPa afterward (Figure 9c). We assess this increase to be significant given that the three tests before MzeroA all ranged between 16–16.8 MPa and the three tests after MzeroA (and before MzeroB) ranged between 16.9 and 17.6 MPa. Jacking pressure values reported by Bröker, Ma, Gholizadeh Doonechaly, et al. (2024) were 13.3–15.3 MPa, that is, lower than all values of our SPT tests. The result indicates that stress conditions in the fracture network have changed over the course of interval 11 stimulations, and also due to the MzeroA stimulation.

As already pointed out in Figure 3c, the number of seismic events induced during the seven SPT tests (Figures 8c and 9d) varies substantially between the different tests. In addition to the number of events, the onset of seismicity, or the seismic reactivation pressure, changes for the SPTs. The pressure at which the first 10 events have occurred here serves as a proxy for the seismic reactivation pressure (Figure 9e). It tends to be higher than the jacking pressure before MzeroA. After MzeroA, the seismic reactivation pressure is well below jacking pressure. Again, the seismic reactivation pressure already dropped slightly immediately after MzeroA (SPT4) but further decreased in the later tests (SPT5–7) and reached low values of 10–11 MPa. It is also noteworthy that during SPT7

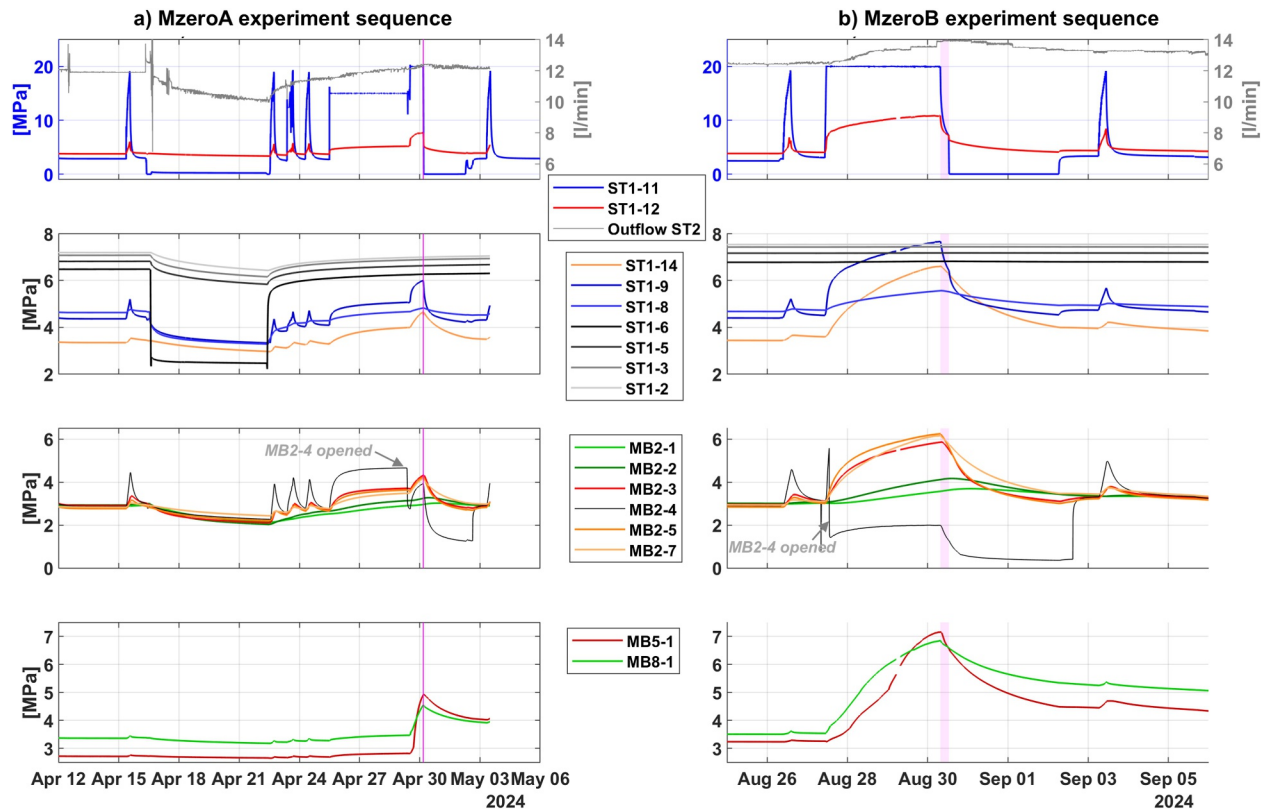


Figure 10. Pressure responses during (a) Mzero experiment sequence and (b) MzeroB experiment sequence. First row: injection pressure (uphole) and interval pressure (downhole) in Interval 12. Second row: Interval pressure (downhole) of all other intervals in ST1. Note that the downhole sensors in interval 4, 7, 10, and 13 in ST1 have not been functional since the installation. Third row: Interval pressure at the monitoring borehole MB2. Note that these intervals are monitored with uphole sensors. Fourth row: Pressure responses of short monitoring intervals in MB5 (i.e., MB5-1) and MB8 (i.e., MB8-1).

(after MzeroB), seismicity was very responsive to pressure increases; the seismicity rate increased each time the pressure was increased at high pressure levels Figure 9c).

5.3. Pressure Responses

Pressure responses observed in intervals of ST1 and the surrounding pressure monitoring intervals in MB2, MB5 and MB8 are shown in Figure 10. Extended time series of monitoring locations are in Figure S6 of Supporting Information S1. Generally, the pressure changes during MzeroB are larger than MzeroA, owing to the much larger injected volume. The intervals of ST1 contain downhole sensors (Figure 10, first and second rows). The pressure responses in Intervals 9 and 12, and to lesser degree intervals 8 and 14, indicate that there is a hydraulic connection to the injection interval 11. A hydraulic connection to interval 12 corroborates the observation that seismicity patterns also align with fractures intersecting interval 12 (Figure 4, Figure S4 in Supporting Information S1) and the observed temperature disturbance (Figure 7). (Note that the pressure sensor in interval 10 does not work, and a similar assessment cannot be given.) In contrast, the deeper intervals 2, 3, 5 and 6 hardly respond to the stimulation. However, they clearly respond to the production test (16–22 April), during which intervals 6 and 7 were opened. These intervals below interval 7 belong to a distinct hydraulic compartment with little connection to the upper intervals (see also Figure S7 in Supporting Information S1).

The pressure in the intervals of MB2 (Figure 10, third row) shows a clear pressure response to the stimulations. The smallest response was in the deepest intervals of MB2-1 and MB2-2. Interval 4 in MB2 (labeled MB2-4) has the strongest connection to the injection interval. However, because of the strong pressure increase, the interval was opened during the main stimulation to prevent rupture of the uphole plastic lines.

Compared to the intervals of MB2 and ST1, the pressure responses of the two intervals MB5-1 and MB8-1 show higher complexity (Figure 10, fourth row). These intervals are short, only 1 m long cells with a downhole pressure

sensor (so-called CTPP sensors, Plenkers et al., 2023). The pressure changes in these intervals are only small during the injections before preconditioning and during preconditioning. For instance, in MB5-1 pressure only slightly increases from 2.7 to 2.8 MPa during preconditioning. About 4 hr after the start of the MzeroA main stimulation, the pressure increases suddenly, leading to a maximum pressure of 4.9 MPa at the end of injection. Similarly, during the stimulation of MzeroB, two episodes of sudden changes in pressure rate leads to a maximum pressure of 7.1 MPa at shut-in.

Before MzeroA, the hydraulic connection between the injection interval 11 and these other intervals was small; however, the main stimulations evidently lead to pressure breakthroughs that connect interval 11 to MB5-1 and MB8-1 hydraulically. Comparing the location of these monitoring intervals with the seismicity cloud during MzeroB experiment (Figure 11b) shows that these newly established connections are formed in the absence of nearby seismic events. MB8-1 is closer to the seismicity cloud than MB5-1, but is still at 10 m distance. The situation is different in interval 12; here, pressure increasing to 7.7 MPa (MzeroA) and 10.8 MPa (MzeroB) were associated with the occurrence of seismicity near the interval.

5.4. Deformation Responses

Time series of selected FBG sensors shows a diverse range of deformation responses (Figure 12, Figure S8 in Supporting Information S1 for the complete set of FBG sensors across the entire experimental sequences). Many sensors show clear extensions (MB8-2-05, MB1-2-04 and MB5-2-06 in Figures 12c and 12d). This indicates a hydraulic connection from the injection interval via the fracture network: so that pressure migration through the fractures causes extension at these sensors. Deformation at these sensors reacts immediately to changes in injection rates: extension starts as soon as injection starts and ceases when injection is stopped. We classify this type of immediate response as **Category I response**. Typically, the response occurs at sensors relatively close to injection (Figure 11), that is, at distances <23 m during MzeroB. These sensors show clear extensional deformation of >10 μe during the SPT tests (Figures 11a and 11b).

A different type of response is observed at most sensors (e.g., Figures 12e and 12f) and is classified as **Category II response**. At the beginning of injection, these sensors first respond with a compression (labeled with (1) in Figure 12f) that peaks at minutes-hours after injection starts. Then, the extension signal starts to take over, reducing the overall compression magnitude (2), until the net deformation at the end of the injection is positive. After shut-in the pattern is reversed; first the sensors respond with further extension up to a peak reached minutes to hours after shut-in (3) followed by reduction of extension. This response is repeated or enhanced after bleed-off. These responses occur at distances >23 m away from the injection.

The **Category III response** includes sensors that respond with compression throughout the experiment and recovery after shut-in (Figures 12e and 12f). These are typically even farther away from injection (>65 m).

As can be seen from Figure 11b, most sensors with Category II responses showed a net extension at the end of the MzeroB stimulation. However, many of these Category II sensors showed only compression up to the end of the SPT tests (Figure 11a). It seems that it depends on the injected volume (MzeroB 147 m³; SPT, 4.4 m³), if the initial compression is overcome by extension and ultimately leads to a net extension (Category II), or if the response remains compressive (Category III).

A fourth type of response (**Category IV**) is only observed for one sensor (MB8-2-03) that is relatively close to injection and is located between two category I sensors that measure strong extension. It shows predominant compression, possibly related to pressurization and opening of neighboring transmissive fractures, which in turn compress the volume where the sensor is located. Also at this sensor, the first phase of intense compression is followed by an extensional trend, reducing the overall amplitude of compression, and leading to a net extension once the injection is stopped.

Often, the long-term extension (or compression) trends during the experiments are interrupted by episodes of sudden extensions or shortening (steps in time series, e.g. Figure 12c). Typically, they reach magnitudes of a few μe . Extension episodes must be related to hydromechanical processes, such as local fracture opening or slip-related deformation induced by localized deformation on a nearby fault or fracture.

Such sudden deformation episodes are prominently observed on sensor MB5-1-04, and to a lesser degree also MB5-2-03 (Figures 12g and 12h). These sensors belong to Category II and are located only 1.75 m from each

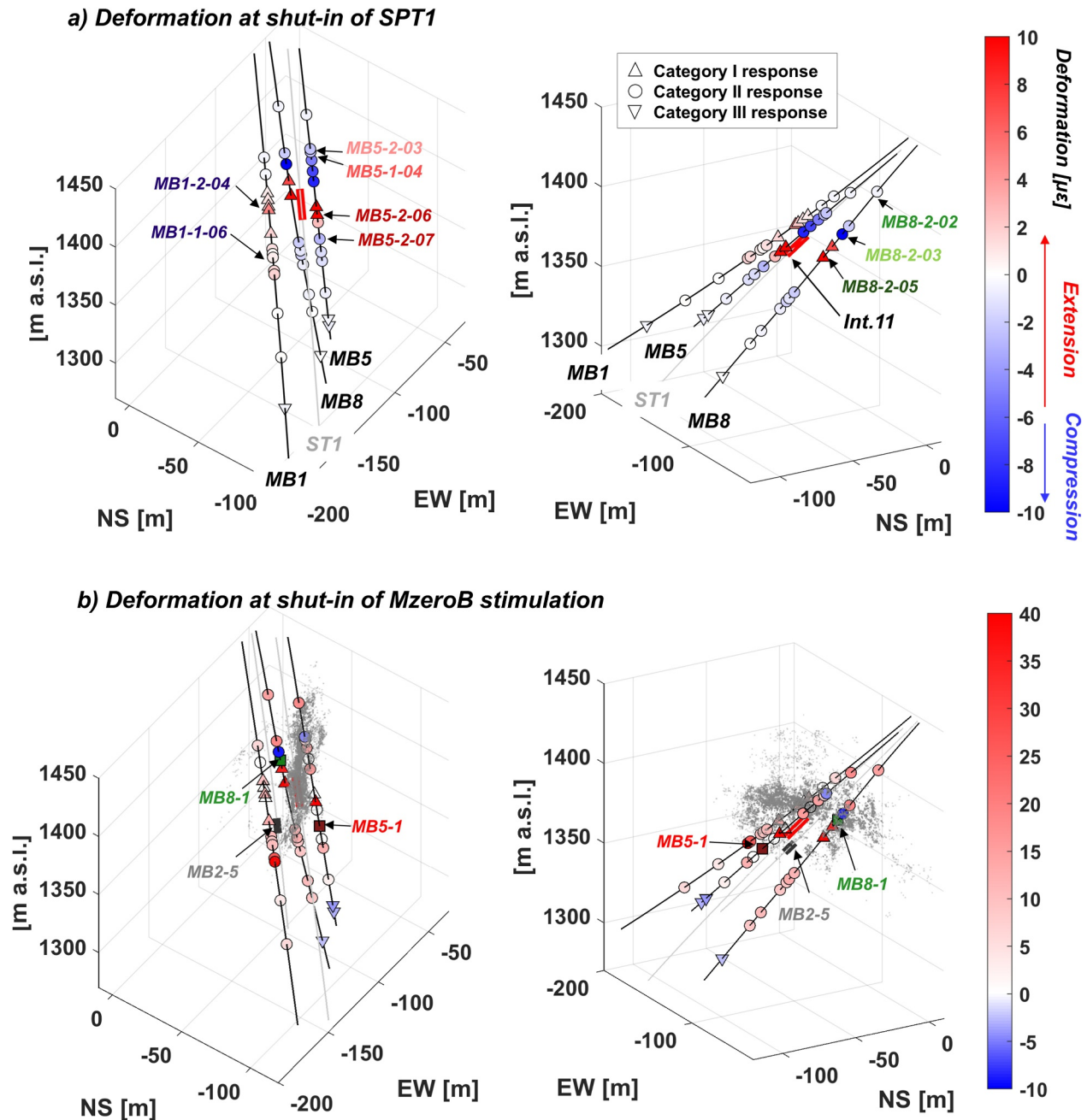


Figure 11. (a) Monitoring boreholes MB1, MB5 and MB8, which contain FBG sensors. The FBG sensors are shown as different symbols highlighting the different deformation response categories. The color of the symbols corresponds to the accumulated deformation during the SPT1 injection, that is, the deformation value at the beginning of the experiment was subtracted from the deformation value just before shut-in. (b) The same for the deformation accumulated during the MzeroB experiment and, for comparison, of the first HTPF test at the beginning of the Mzero experiment sequence. Also included are the pressure monitoring locations MB5-1, MB8-1 and MB2-5.

other. During MzeroA, they follow a nearly synchronous trend (i.e., they overlap in Figure 12g). They are also synchronized during the first hours of the MzeroB main stimulation, and start differing more strongly after few hours of stimulation (Figure 12h). This separation coincides with the time when the seismicity cloud reaches these sensors. Seismicity has recorded as close as 2 m from the sensors around the time that the time series begin to separate. More seismicity is located close to MB5-2-03, the sensor with fewer sudden shortening/opening episodes, than to MB5-1-04.

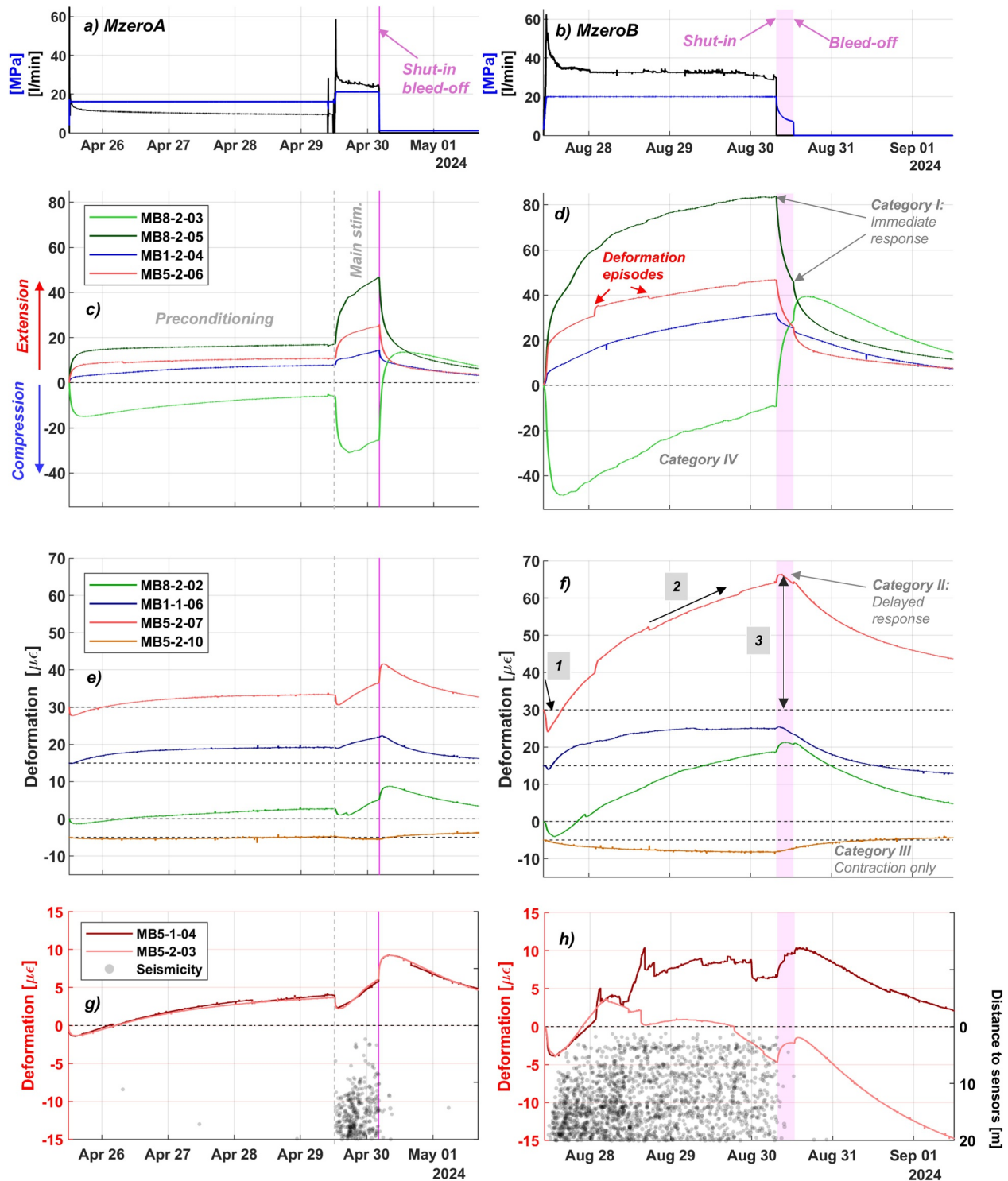


Figure 12. Representative FBG deformation time series of MzeroA (left column) and MzeroB (right column). (a, b) Injection parameters of MzeroA and MzeroB, respectively. (c, d) Deformation time series with an immediate drop in extension at shut-in or bleed-off. (e, f) Deformation time series with a delayed response (i.e., further extension) to shut-in/bleed-off. (g, h) Two sensors that were hit by the seismicity cloud during MzeroB (i.e., the sensors are within the seismicity cloud). Gray dots denote seismic events for which the distance to the two sensors is indicated (right axis).

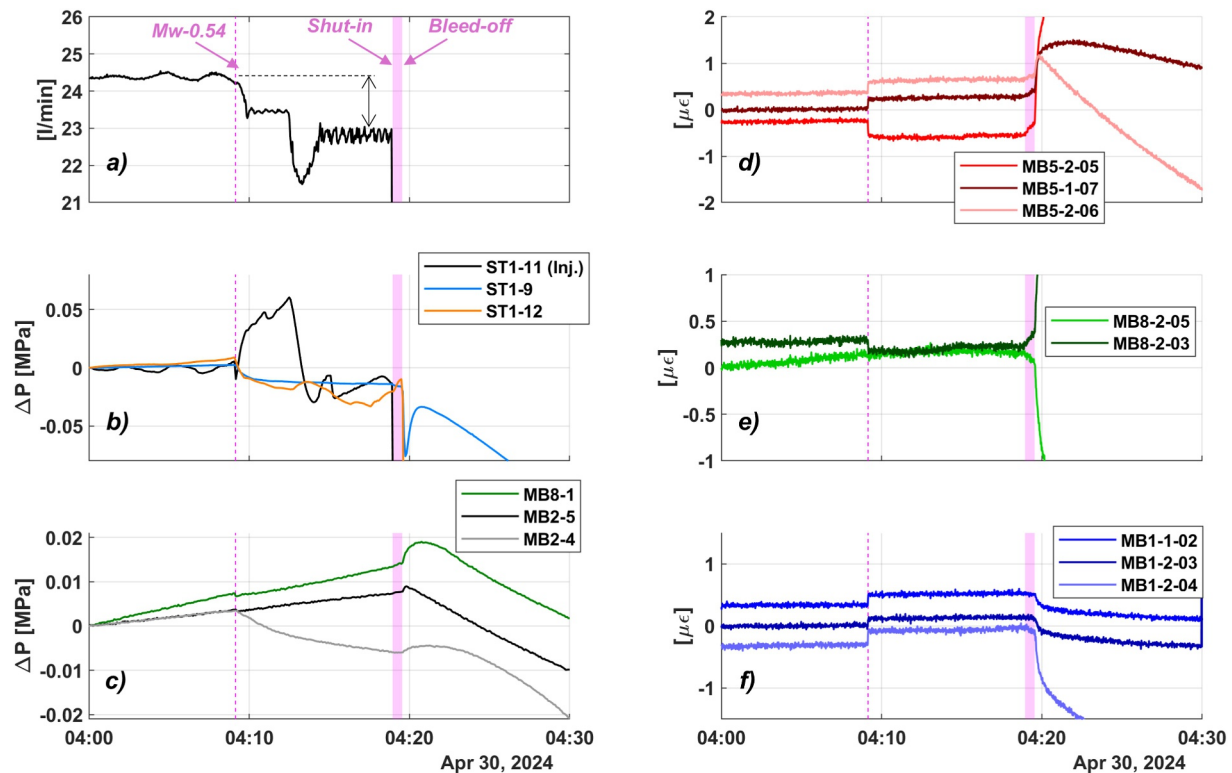


Figure 13. Co-seismic responses to the Mw-0.54 event and responses to shut-in and bleed-off of the well. (a) Injection rate. (b) Pressure change referenced to beginning of presented time interval of injection interval (ST1-11) and intervals ST1-9 and ST1-12. (c) Pressure changes of monitoring intervals MB8-1 (CTPP sensor) and MB2-4 and MB2-5. (d–f) Deformation at selected FBG sensors in boreholes MB1, MB5 and MB8.

Moreover, most of the extension/shortening episodes could not be associated with any detected seismic events. Also, many other sensors show more sparsely occurring deformation episodes, and these are likewise not associated with seismic events. We acknowledge that the local details of deformation represented by these episodes remain unknown. Also, we cannot fully rule out the possibility that they may reflect cracking of the borehole grout, in which the sensors are embedded. Nevertheless, we conjecture that they must be associated with fluid-activated deformation in the vicinity that appears to be mostly aseismic. Most deformation may occur at accelerated but nonetheless sub-seismic rates.

Comparing the deformation before and after the experimental sequences (i.e., at the beginning and end of the timeseries in Figure S8 of Supporting Information S1), we can infer the permanent total deformation that is the sum of seismic and aseismic deformation. For MzeroA, this ranges from -3 to $8 \mu\epsilon$, for MzeroB from -22 to $18 \mu\epsilon$. The spatial distribution of permanent deformation shows complexity potentially owing to the short-term local deformation effects (Figure S9 in Supporting Information S1) and will be the subject of future studies.

5.5. Co-Seismic Hydromechanical Effects

The signature of the Mw-0.54 mainshock of the MzeroA experiment is clearly seen on various deformation and pressure sensors (Figure 13). About 3 min after the occurrence of the mainshock, flowrate dropped from about 24.5 l/min to about 23 l/min (Figure 13a). While transient phases of increased flowrate have occurred several times, a comparable decrease in flow rate has not been observed throughout the main stimulation. The delay between the mainshock and the flowrate drop is related to the pump circuit having a delay in responding to changed conditions. Thus, we assess the decrease in flow rate to be associated with the MzeroA mainshock. The pressure in the injection interval (i.e., labeled ST1-11 in Figure 13b) shows why the pressure-controlled pumps had to respond in an attempt to maintain the 20 MPa injection pressure: when the mainshock occurred a short drop in pressure occurred, that was likely associated with a local volumetric expansion of the interval. Then the pressure increased, indicating that the resistance to flow increased in the near-wellbore fracture network. The

Table 1
Summary of Differences Between MzeroA and MzeroB Stimulation

	MzeroA	MzeroB
Transmissivity before/after (m ² /s)	7.3e−7 3.6e−7	4.7e−7 9.9e−7
Injectivity before/after (l/min/MPa)	0.94 1.05	1.04 1.40
Jacking pressure before/after (MPa)	16.4 ± 0.21 17.9 ± 0.28	17.1 ± 0.13 17.6 ± 0.27
Ambient pressure before (Int. 11/Int. 2) (MPa)	3.94 7.23	4.09 7.54
Steady-state flow rate at 20 MPa (l/min)	24	33
Injection duration (hh:mm:ss)	16:00:57	69:36:25
Injected volume (m ³)	24.6	146.5
Number of seismic events above −4.25 during injection*	3,960	31,058
Seismic event rate per time* (1/hour)	243	447
Seismic event rate per volume* (1/m ³)	161	212
<i>b</i> -value	1.31 ± 0.02	1.17 ± 0.01
Seismogenic index	−3.36	−2.15
Magnitude of completeness, Mc	−4.34	−4.27
Largest magnitude Mw	−0.54	−0.51
Largest magnitude jump ΔMw	1.02	0.40

Note. For MzeroA only the outcomes of the main stimulation, that is, excluding the preconditioning phase, is reported. Seismic event rates only consider events with magnitude above the magnitude of completeness, that is, MwA > −4.25. Ambient pressure values are taken from target interval 11 and interval 2 in borehole ST1 on 18 March 2024 and 15 August 2024, that is, at times of steady state pressure conditions before the experiment sequences. The largest magnitude jump denotes the difference between the largest and the second largest event. *Seismic event numbers and rates only consider magnitude above a conservative magnitude of completeness of Mc = −4.25, which is higher than the individual Mc per experiment. The shut-in period is excluded, that is, only event during injection are considered.

pumps adjusted to lower flow rates to maintain the uphole injection pressure, so that the interval pressure decreased. The intervals 9 (ST1-9) and 12 (ST1-12) adjacent to interval 11 similarly show a decrease in pressure after the mainshock, which may be associated with a compromised connection to the injection interval. Also, the pressure monitoring intervals MB8-1 and MB2-4 reacted to the mainshock (Figure 13c) with a small pressure drop (MB8-1) and gradual increase of pressure starting the moment the mainshock occurs (MB2-4). The latter may be related to a compromised hydraulic connection between monitoring and injection intervals. The changes in the hydraulic conditions inferred from the pressure and flowrate evolution in Figure 13 are in agreement with a drop in transmissivity observed before and after the stimulation (Figure 6).

Several FBG sensors show small but step-like changes in deformation of up to 0.3 με, that occurs concomitantly with the mainshock (Figures 13d–13f). Permanent deformation of up to 0.8 με has also been recorded on DAS cables in boreholes MB3, MB5 and MB8 (see Figure S10 in Supporting Information S1). The general patterns observed with FBG and DAS agree despite some discrepancies, which may be related to artifacts in the measurement technology or the installation of the sensors and sensing cables and are currently being investigated.

6. Discussion

We tested two distinct hydraulic stimulation protocols—with and without hydraulic preconditioning—to experimentally evaluate if preconditioning can enhance the chance of larger induced earthquakes. Indeed, the two experiments result in distinct characteristics in terms of seismicity rate, spatial distribution, and magnitudes, but also in the hydraulic and hydromechanical response (Table 1). Average events per time and per injected volume was higher during MzeroB compared to MzeroA (Figure 3), especially during early times. The

propagation style of the seismicity cloud differed between the experiments (Figure 4). During MzeroA, there was a step in magnitude between the main shock and the seismicity before. This mainshock produced co-seismic effects clearly seen on the pressure and deformation evolution (Figure 13). In contrast, MzeroB did not have such a distinct large mainshock but several relatively large events. The two experiments also differed in terms of hydromechanical effects. Transmissivity decreased after MzeroA, while after MzeroB it increased (Figure 6). During stimulation with 20 MPa, the injection rate was higher during MzeroB than MzeroA (Figure 2). There was little change in injectivity from SPTs after MzeroA, but injectivity increased after MzeroB (Figure 8). The deformation time series reveals hydromechanical responses to the different injection schemes (Figure 12).

In the following, we discuss whether the preconditioning scheme is responsible for the MzeroA mainshock and the different seismicity characteristics of MzeroA and MzeroB, or if changes in the rock mass between the experiments produced these different outcomes. To do so, we need not only to consider the full range of hydromechanical effects, but we must also set the two experiments into context with the entire stimulation history in interval 11 (Figure 5).

The SPT injections show that the seismicity rate increased after MzeroA and seismicity occurs at lower pressures with each SPT (Figure 8). While this increasing seismogenic response corroborates with the more intense seismic activity during MzeroB, it is contrary to what is expected. The “Kaiser effect”, often observed in hydraulic stimulations (i.e., Kaiser, 1953; Li & Nordlund, 1993; Zang et al., 2021), predicts that seismicity decreases with each experiment because of stress relaxation. From the past stimulation in the same interval (Figure 5a), the Kaiser effect can indeed be inferred, because seismicity rates decreased more than an order of magnitude up until the beginning of the Mzero sequences. Thus, it is curious that after MzeroA seismicity rate and spatial characteristics emerged as if it had been “reset” so that it appears as if the fracture network had not seen any stimulation before. The seismicity rate of MzeroB has been larger than during MzeroA, which is the opposite to the expectation that the fracture network has been destressed by MzeroA. Obviously, there were changes in the fracture network between experiments that made it progressively easier to seismically reactivate it. Possible explanations may include:

- Co-seismic stress redistribution related to the Mw–0.54 mainshock;
- Time-dependent aseismic processes accommodating the stress redistribution related to seismic and aseismic slip along fractures.
- Change in ambient pore pressure between MzeroA and MzeroB;

In the following, we elaborate on these three processes and discuss how they impact seismicity in addition to the applied hydraulic preconditioning scheme.

6.1. Co-Seismic Stress Redistribution

Co-seismic stress redistribution due to the Mw–0.54 mainshock is evident from the deformation and the subtle changes in the hydraulic connections (Figure 13). Using the observed co-seismic deformation, a model of the mainshock source area, orientation and slip magnitude could be calibrated (Text S9 in Supporting Information S1). The inferred source orientation and slip direction (oblique normal faulting) agreed in principle with the focal mechanism from first motion polarity analysis (Meier et al., in prep). Forward-modeled stress redistribution from this source model (Figures 14b and 14c) predicts that the stress change associated with the deformation in Figures 13d–13f is on the order of 1–10 kPa (i.e., normal stress component in borehole direction). Similarly, the model predicts volumetric stress changes of ~6 kPa in the injection interval 11 and about ~4 kPa in interval 9, which is in reasonable agreement with the sudden pressure drops associated with the main shock (Figure 13b). These stress changes around the injection interval may be responsible for the drop in transmissivity Figure 6, the changes in flow regime (Figure 7 and Figure S4 in Supporting Information S1) and changes in connectivity between injection and monitoring intervals after MzeroA. Note that similar observation have been made in other hydraulic stimulation (e.g., Krietsch et al., 2020). They also have implications for hydraulic stimulations of geothermal reservoirs; decreases in transmissivity and changes in the hydraulic system related to larger seismic events would be averse to the operational goal of enhancing transmissivity.

The model has further been used to compute changes in differential stress at the seismic event locations of the MzeroB experiment (Figure 14). An increase in differential stress changes associated with the main shock exceeds 0.1 MPa only within 5–10 m of the main shock. However, most MzeroB seismicity—in particular at early

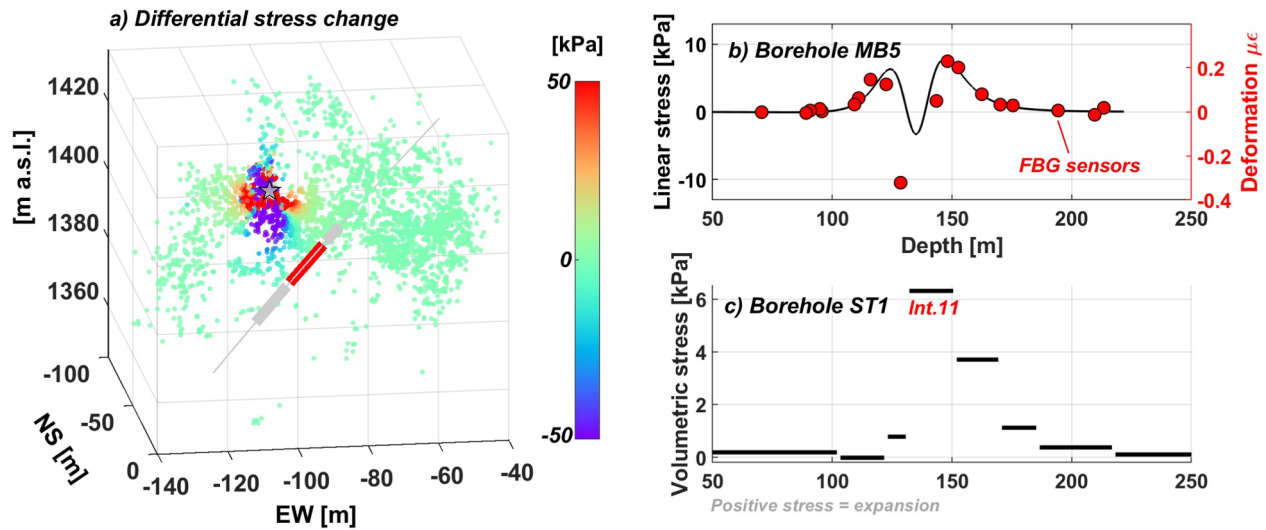


Figure 14. (a) Differential stress change associated with the Mw−0.54 that occurred at the end of the MzeroA main stimulation. Differential stress change is resolved on the MzeroB seismic event locations. (b) Linear stress changes in borehole MB5 induced by the main shock resolved compared to the FBG sensor locations. (c) Volumetric stress changes at the ST1 intervals. Note that positive linear stress changes and volumetric stress changes denote extension.

times—occurred at locations where stress change was <0.1 MPa (Figure 14). Thus, locally around the main shock, seismicity may have been promoted by stress redistribution, which explains why the SPTs after MzeroA produced seismicity in this region (Figure 5) and at comparably lower pressure levels than before MzeroA (Figure 9c). Note that these estimates of stress interactions are lower bounds, as the remaining large number of seismic events may also contribute to the stress changes by MzeroA. Furthermore, the larger seismic events of MzeroB may be associated with similar stress interactions as evidenced by the short-lived seismicity rate increases after the Mw−0.95 and −0.91 during MzeroB (Figure 3b).

6.2. Aseismic Deformation

In addition to seismically induced stress transfer, aseismic slip may also contribute to stress changes in the fracture network around the interval (e.g., Cappa et al., 2019). Various authors demonstrate that aseismic slip during or after injection plays a key role in understanding induced seismicity (Bourouis & Bernard, 2007; Guglielmi et al., 2015; Scotti & Cornet, 1994; etc.), and it is well known that seismic slip may only be a small percentage of total slip (De Barros et al., 2019; Duboeuf et al., 2017; Villiger et al., 2020). Sáez and Lecampion (2023) suggest that aseismic stress redistribution may be equally, if not more, important than stress redistribution produced by seismic events in explaining seismicity after hydraulic stimulations. Deformation episodes that are not associated with seismic events (Figure 12) of a few $\mu\epsilon$ and total deformation (i.e., seismic + aseismic deformation; Figures S8 and S9 in Supporting Information S1) measured after the experimental sequences is on the order of 1–10 $\mu\epsilon$. Expanding on the stress change estimates associated with 0.1–1 $\mu\epsilon$ co-seismic deformation, this would imply 10–100 kPa of stress changes associated with aseismic deformation episodes (Figure 12) and the total permanent deformation (Figure S9 in Supporting Information S1). Thus, total stress redistribution associated with all seismicity and aseismic slip is orders of magnitude larger than predicted from mainshock alone. The increase in jacking pressure after MzeroA (Figure 8) of >1 MPa may be further evidence of stress redistribution.

6.3. Ambient Pressure

A further process evidenced in our data is an increase in the ambient pressure between MzeroA and MzeroB. In the injection intervals, the difference in pressure prior to both experiments was about 0.1 MPa. The ambient pressure evolution shows a clear seasonal change with up to 0.3 MPa peak-to-peak amplitude (Figure S6 in Supporting Information S1). Superimposed to this seasonal signal is a comparably strong pressure decrease of 0.7 MPa induced by the production test. If we use a distant interval (Interval 2 in borehole ST1) as a proxy for the ambient pressure level that is met by the stimulation, we find that it correlates well with the varying number of seismic events of the SPT experiments (Figure 9d). This would imply that a seasonal pressure increase induced by

snowmelt may similarly lead to a preconditioning of the fracture network thus modifying the seismogenic response. Recently, Simon et al. (2025) found that seasonally appearing seismicity clusters could be associated with snowmelt and glacier melt infiltration. Seismicity modulated by seasonal hydrological mechanisms has been observed in many cases world-wide and is well-documented in literature (e.g., Bettinelli et al., 2008; Ueda et al., 2024; Zhao et al., 2025). Note that the magnitude of pressure changes measured here is in the same order of magnitude as those discussed coming from seismically and aseismically induced stress interactions. However, compared to these effects, the spatial footprint of these ambient pressure changes is much larger. Thus, the contribution of these pressure changes in altering the seismogenic response is plausible. Repeating standardized stimulations over the course of an entire seasonal cycle at the BedrettoLab might allow further investigating these hypothesized seasonal effects.

6.4. Poroelastic Effects of Preconditioning

While seismic and aseismic stress interaction as well as ambient pressure changes may impact the seismogenic responses, we also assert an important role to the different injection protocols in producing different seismicity characteristics of MzeroA and MzeroB. Modeling studies indicate that a faster increase of pressure may lead to more and larger seismic events (Almakari et al., 2019; Rudnicki & Zhan, 2020), which would explain the outcomes of MzeroB. In contrast, other studies (e.g., Ciardo & Rinaldi, 2022; Langenbruch et al., 2024; Moein et al., 2026) indicate that slow increase of pressure may produce larger ruptures. This supports that extended preconditioning before MzeroA may have played a role in facilitating the MzeroA mainshock, which seems to be a step-out event of the sequence.

A general mechanism of how preconditioning enhances ruptures may be interpreted from the deformation responses presented in Figure 12. The different deformation categories can be interpreted as poroelastic responses, that differ depending on distance from injection and time elapsed since start of injection as sketched in Figure 15. Upon start of the injection, the fluid pressure increases almost immediately in the connected fracture network close to the injection interval. This may be referred to as a “drained response.” Thus, sensors in this area immediately extend and, once injection halts, immediately contract (Category I). Further away from injection, radial compression and tangential extension occur initially, because of the increase of fluid volume and pressure close to the injection. This corresponds to an “undrained response.” Sensors in this area oriented in a near-radial direction show compression in the beginning. Once the fluid and pressure front further migrates to reach these sensors, deformation turns around to become more extensional with time. This drained response is superimposed to the undrained response. If injection continues for sufficiently long, deformation becomes net positive, that is, extensional at the end of injection. The effect is reversed once injection stops; poroelastic extension related to the pressure dissipation close to injection interval leads to enhanced extension. This is followed by compression, once the pressure decrease at the sensor itself starts dominating over the poroelastic extension. When the borehole is opened (bleed-off), not only does the pressure dissipate, but fluid volume is effectively reduced around the injection interval leading to additional extension farther away. Typically, this delayed response with initial compression followed by extension and a post-shut-in extension occurs farther away from injection, that is, 18–95 m.

These poroelastic effects have been described by various authors as a key mechanism underlying induced seismicity (Boyet et al., 2023; Deng et al., 2016; Segall & Lu, 2015; Zhai et al., 2019). However, direct measurements of them from within the rock mass have rarely been documented as done here (e.g., Dutler et al., 2021). These poroelastic effects influence stress conditions along fractures and therefore rupture propagation upon reactivation. The radial poroelastic compression beyond the pressurized fracture network at early times may lead to local clamping (i.e., increase of total stresses) of the fractures of certain orientations. Once fluid and pressure migrate to these fractures, the extension signifies that the effective stress decreases thus promoting slip. Although locally the fracture orientation in connection with these transient poroelastic effects adds complexity, seismicity would preferably occur at locations with increasing extension, the example in Figure 12h serving as example.

If the fracture network is not preconditioned, that is, if pressure is directly ramped up to high pressure, there is a compressive and clamping “cage” around the pressure front, beyond which slip is counteracted. In contrast, if the fracture network is preconditioned, there is an extended rock volume in which the effective normal stresses are reduced through fluid pressure, while staying below reactivation pressure. If pressure is then ramped up beyond the reactivation pressure, the poroelastic compression and clamping still occur, but the net effective normal stress

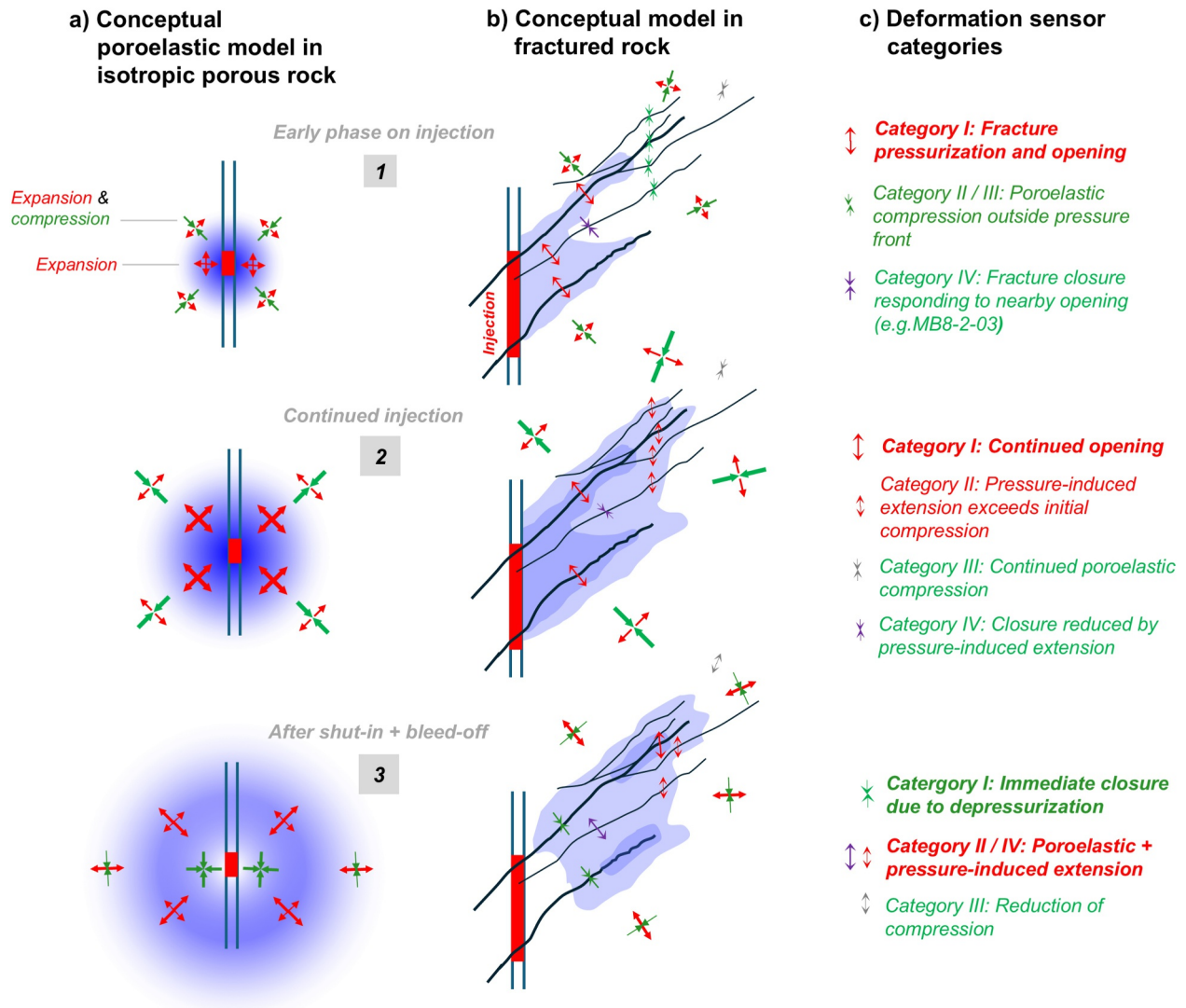


Figure 15. Sketch of different poroelastic effects during the phases highlighted in Figure 12. (a) Conceptual model of poroelastic field in a homogenous isotropic medium. (b) Conceptual of poroelastic effect in fracture rock. (c) Explanation on behavior of different deformation categories during different phases of the injections.

is still reduced. This is evident in Figure 11: despite a compressional signal after changing from the preconditioning pressure of 15 MPa to the main stimulation pressure of 20 MPa, the net deformation is extension. There are two reasons for this: (a) The compressional signal is less pronounced compared to the one at the beginning of MzeroB, because the pressure increase is only 5 MPa, as compared to ~16.5 MPa when going from ambient pore pressure to 20 MPa during MzeroB. (b) Through preconditioning the accumulated extension exceeds the early time's compression signal. Many fractures are thus already prepared to rupture, and rupture propagation may be less inhibited by poroelastic clamping at the margin of the pressurized volume. This might explain why seismicity is more evenly distributed in the seismicity cloud (Figure 4) and potentially also why a larger rupture such as the step-out Mw-0.54 mainshock was possible. Instead, during MzeroB, the separation between the pressure front and the poroelastic compression ahead of it is more pronounced so that seismicity follows a clearer propagation pattern.

Note that a change in ambient pressure may be regarded as a natural preconditioning process encompassing rock mass on the mountain-scale. Clearly, our preconditioning protocol acted on a smaller rock volume but at orders of magnitude larger pressure levels (>10 MPa instead of >0.1 MPa). Nevertheless, it is noteworthy that comparably small pressure changes may be sufficient to affect seismicity. The different mechanisms discussed here, including the poroelastic compression ahead of the pressure front, operate at stress levels smaller than 1 MPa, but impact the

seismogenic response. Studies of earthquake-earthquake interaction of aftershock sequences (e.g., Martínez-Garzón et al., 2026; Meier et al., 2014; Toda et al., 2005; Ziv & Rubin, 2000) and dynamic triggering of remote earthquakes (Roth et al., 2025) similarly point out that seismicity may be triggered or modulated at stress change levels of <0.1 MPa, even down to 0.001 – 0.01 MPa. Also, the observations in Figure 12h show that the deformation (and pressure changes) ahead of the seismicity front is surprisingly small. If seismicity followed a larger pressure increase, the deformation before seismicity arrives would be orders of magnitude larger than observed at the deformation sensors in Figure 12h.

7. Conclusions

Much research on induced seismicity strives for injection strategies that reduce seismicity. Many of these concepts have been implemented in large-scale hydraulic stimulations for geothermal reservoir development (e.g., Pohang, Helsinki; Hofmann et al., 2018, 2019; Kwiątek et al., 2019). We went in the opposite direction to find a stimulation procedure that enhances seismicity aiming at an induced $M_w \sim 0.0$ earthquake in an underground laboratory to be studied in great detail with a multi-disciplinary 3D monitoring network. For this, we tested an injection procedure, in which the fracture network was pressurized at pressure levels below fracture reactivation pressure for several days, to precondition it for seismic rupture during stimulation above reactivation pressure (i.e., MzeroA). For comparison, we also conducted a multiday stimulation that was started by quickly ramping up pressure (i.e., MzeroB), that is, without any intentional pre-conditioning. The experimental outcomes were successful in the sense that a step-out $M_w - 0.54$ earthquake (initial magnitude estimate $ML_c - 0.2$) was induced within the dense monitoring network during MzeroA. The earthquake had a magnitude more than one level larger than the second largest preceding earthquake and exhibited an extraordinary aftershock sequence. However, The MzeroB experiment without preconditioning several months later led to larger event rates and several stronger events $M_w - 0.9$ to -0.5 events, albeit without step-out magnitudes.

We analyze the induced seismicity history of the same borehole interval along with hydromechanical observables within the fractured rock mass to assess a potential seismicity-enhancing role of preconditioning from these experimental outcomes. The analysis remains inconclusive, because we are confronted with common difficulties in experimentally evaluating stimulation strategies. These arise from the fact that the hydromechanical conditions in rock mass (reactivation stresses, transmissivity, flow paths, ambient pressure conditions, etc.) change for each subsequent experiment and also influence the experimental outcomes in addition to the stimulation strategy. Indeed, we present quantitative observations of stress relaxation and stress transfer of seismic slip and aseismic processes as well as of the ambient pressure condition undergoing a seasonal variation, which likely worked in concert to alter the seismogenic response of the fracture network. These effects are not limited to induced seismicity but may similarly condition natural earthquake sequences. In the context of induced seismicity, a temporally changing seismogenic response of the same fracture network poses challenges for seismic hazard forecasting and management. Designing and testing injection protocols that minimize (or in our case maximize) seismicity have to consider the additional complexity of changing experimental conditions in subsequent injection tests.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Data files used in this publication are available at (Gischig et al., 2026), which includes injection flow rate, pressure and deformation data relevant for the Mzero experiments A and B. Seismicity catalogues referred in this paper are made available by Scarabello et al. (2025) and Roskopf et al. (2024). Data files of the hydraulic stimulation from the VALTER experiment (Phase 1 and Phase 2) are also available from Roskopf et al. (2024).

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Acknowledgments

In the “Bedretto Underground Laboratory for Geosciences and Geoenergies (BedrettoLab),” ETH Zurich studies in close collaboration with national and international partners techniques and procedures for a safe, efficient, and sustainable use of geothermal heat and questions related to earthquake physics. The BedrettoLab is financed by the Werner Siemens Foundation, ETH Zürich and the Swiss National Science Foundation (10.003.096, 200021_192151). The BedrettoLab would like to thank Matterhorn Gotthard Infrastructure for providing access to the tunnel. This project has received funding from the European Research Council (ERC) project FEAR (Grant 856559) under the European Union's Horizon 2020 research and innovation program. Open access publishing facilitated by Eidgenössische Technische Hochschule Zurich, as part of the Wiley - Eidgenössische Technische Hochschule Zurich agreement via the Consortium Of Swiss Academic Libraries.

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